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Claremont McKenna College

NFL Positional Value In Terms of Salary Cap

Spending: Why Are Running Backs

Undervalued?

submitted to
Eric Hughson

by
Charlton Jackson

For
Senior Thesis
Spring 2024
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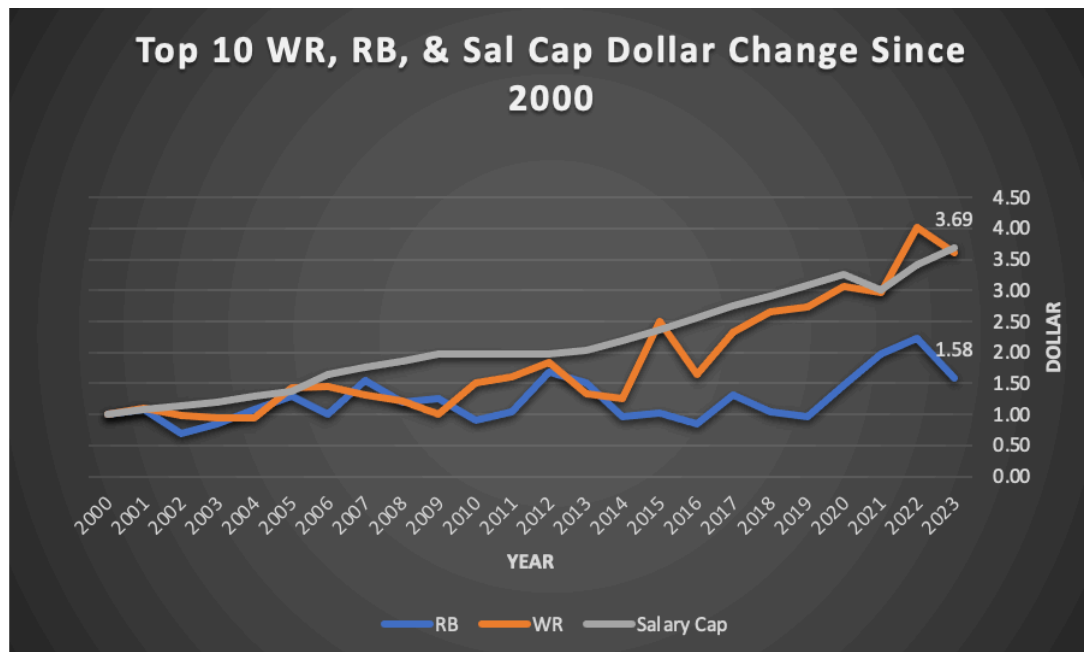
Abstract

Running Backs over the past two decades have seen a contract value devaluation relative to other positions and the league's salary cap spending limit. The growth of running back salaries has stagnated in the past decade, and they currently fail to keep pace with the contracts for other positions. Through our empirical research, we were able to either reject or better explain existing reasoning for recent running back devaluation, such as the increase in passing volume, durability concerns, and running backs by committee. We found that the most significant change in the past 30 years of the NFL, since the 1994 implementation of the salary cap, was the 2011 change in the league's collective bargaining agreement. This rule took all negotiating leverage out of the hands of rookies entering the league and instead placed all drafted rookies on mandatory 4-year, cost-controlled contracts. In doing so, the league's outlook completely shifted from a statistical and roster construction standpoint. Most notably, by altering the rookie contract structures, running backs have been harmed the most. The lion-share of an already shorter career, relative to other positions, now resides on non-negotiable, underpaying contracts when compared to historical norms for the position, how other positions' salaries have grown, and inflation.

Introduction

Since the inception of the NFL, the game has primarily been defined and popularized by its star running backs, who were compensated accordingly. Players such as Jim Thorpe, the league's first household name in the 20s, Jim Brown in the 50s and 60s, Walter Payton in the 80s, and Adrian Peterson in the 2010s were all staples to the NFL brand and saw salaries corresponding to their prestige. Yet in today's NFL, the league's most recognizable faces are quarterbacks, wide receivers, and tight ends. Historically, the salary that top running backs earned was the same as what top receivers earned, but post the 2011 collective bargaining agreement, the running back has fallen behind. Figure 1¹ Illustrates the magnitude of this.

Figure 1



¹ Sorted and downloadable contract data was not available prior to 2011-2013, depending on the size and type of contract information. Thus, to create the chart above, the top ten running backs (in terms of rushing yards) and top ten receivers (in terms of receiving yards) were gathered every year beginning with the 2000 season. Following this, we hand-collected the individual contract information for every player, amounting to a sample size of 480 running backs and receivers, and their respective average yearly salaries for the contract they played on that season.

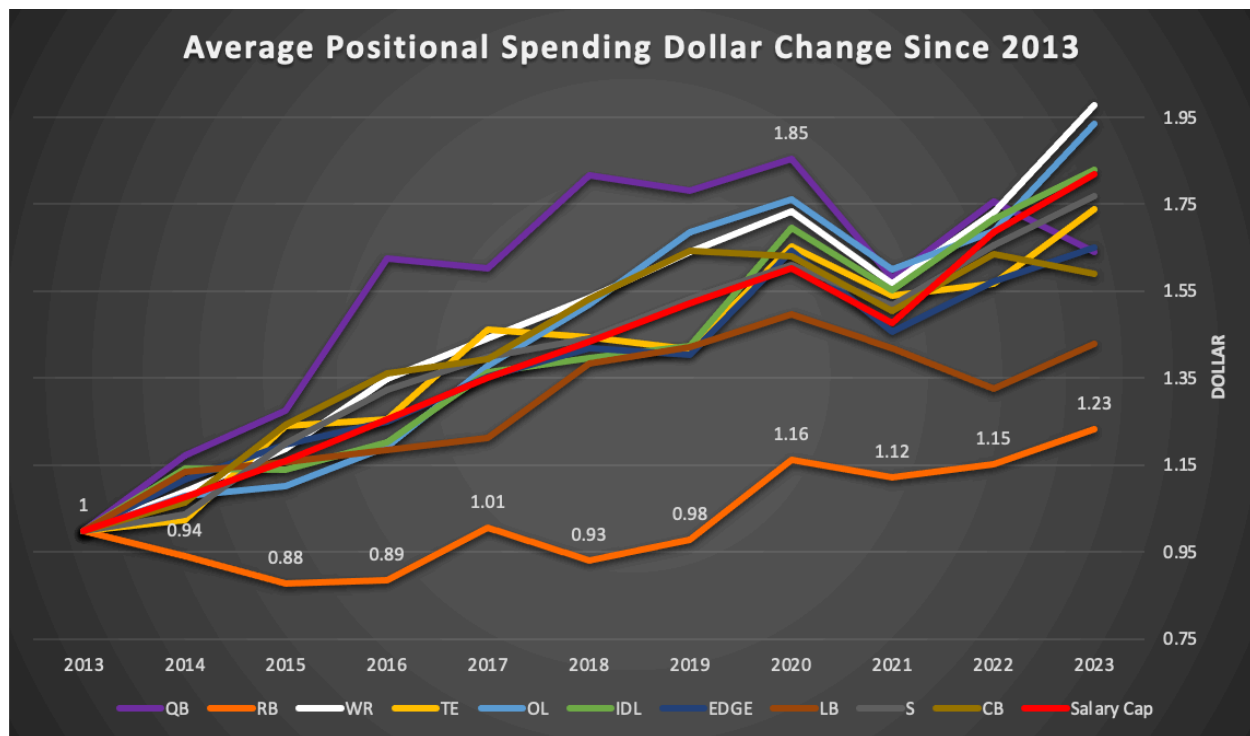
In Figure 1, we normalize the salary cap, the average salary of the top-ten wide receivers, and the average salary of the top-ten running backs to \$1 in the year 2000 and trace each growth from 2000 until the present day. Observe that the salary cap and the average wide receiver salary have increased by 3 ½ times over the period. However, the average salary of top running backs has increased only 1 ½ times. In application, the factoring of inflation, we find these salaries have fallen despite the increased popularity and profitability of the NFL over the last 20 years. The CPI from 2000 until 2023 increased by approximately 70% - slightly more than 2.5% per year.

Unfortunately, amalgamated positional salary data was not available prior to the 2013 season; however, when viewing the NFL average spending by position for this past decade in Figure 2². Again, normalized to a starting \$1 amount like Figure 1, we find running backs to show the least growth by a wide margin. In only the past decade, we have observed running backs dip below our \$1 dollar starting point, which was soon after the 2011 CBA change this research focuses on. Even in more recent years, running backs have only grown about 23%, which lags far behind the near doubling in value of other offensive positions such as OL and WR³. Even with this shortened horizon starting with 2013, running backs trail the change in CPI, which has increased by about 30% since 2013.

² The sample size of Figure two is composed of every rostered NFL player, 54 players per roster with 32 team rosters, and then the league average spending by position group. This figure works similarly to figure 1, except it has a much broader sample size, so it's more representative of average and lower level players rather than just the top ten, has a shorter timeline due to data availability, and has allowed for the comparison with all positions rather than just receivers and running backs.

³ Quarterback contract growth is unique due to how much greater a top quarterback contract extension is than the initial rookie contracts, which are the same for any rookie, regardless of position. This is why Quarterbacks initially showed the most growth and then dropped to mean due to more expensive veterans being replaced by talented rookies who have yet to get their extensions. This, paired with Quarterbacks being the smallest position group in terms of # of quarterbacks rostered, means their growth numbers are very susceptible to fluctuation.

Figure 2



The most common explanations for the devaluation of the running back position are often attributed to shorter careers due to the injury risk of running backs, the advent of a more passing-oriented league, as well as teams favoring building their rushing attack through "running back by committee" where they will use multiple cheaper running backs in favor of one more expensive rusher. Interestingly, these opinions are shared by analysts, current and former players, talk show/media personalities, and even AI alike. For example, on NFL Live and ESPN, sports writer Bill Barnwell cites a move toward passing as a critical contributor to runningback devaluation. Other research papers, which can be found later in the notes section, such as Kennedy (2017), believe the true reason for devaluation to be a matter of early physical deterioration of the running back relative to other positions. Chat GPT believes running back by committee and modern passing are principal reasons for position devaluation. The research and findings of this paper largely reject the previously listed explanations and look to impart a new

interpretation of running back value as well as NFL positional value as a whole. The main driver for running back devaluation is a change in NFL rookie contract law, which was implemented in the 2011 collective bargaining agreement. Not only do we believe this to be the main reason, but we also affirm that all other explanations are merely a byproduct of this 2011 change. This finding is not only verified through the visible contract shifts around this 2011 date, as seen in Figure 1. Contract law as an explanation for devaluation became our answer by eliminating statistical or play style changes as a possible explanation. We found no significant changes to the NFL team's averages over the past 20 years, outside of a sub-10 % increase in completion percentages. On a positional basis, there still has not been much change in how players produce, barring a drop in the numbers of the league's rushing leaders. This supports the notion of running backs by committee, given there are fewer individually prominent rushers but the same in total rushing production, suggesting an increase in the amount of contributing rushers. However, running back by the committee was only made possible through the aforementioned contract structure changes made in 2011.

Before explaining our empirical approach and presenting our data findings, the NFL and its team-building structure must first be explained. These explanations are necessary to understand the research conducted behind this paper and will be referenced throughout the paper.

Technical Explanations

NFL Rosters

During NFL training camp and the preseason, NFL rosters are composed of ninety-man rosters. After the conclusion of the two-week training camp and four-week preseason, the NFL regular season begins, and rosters must be reduced to 53 active players. Sixteen player spots may be signed to teams' practice squads and not count toward the 53-man roster. Players placed on the "injury reserve" will also not count toward the 53-man roster while on the injury reserve. Players on injury reserve must remain on the "IR" for at least weeks before being designated back to the active roster. Teams can designate up to eight players to return from IR per year, but one player may only return twice, counting twice for the eight returns. If a player is placed on the injury reserve prior to the season commencement, they are not eligible to return for the rest of the season. Once a player is designated to return, a 21-day clock begins for them to be activated to the 53-man roster or be permanently returned to IR for the season. Before every game, teams must announce a game-day roster composed of 48 players who are allowed to dress for the game. The five remaining active players will be designated as inactive, which could be for many reasons (injury, ability, etc). Practice squad players may be activated to the game-day roster three times per season before being required to be signed to the 53-man active roster or no longer be available for elevation. Football as a sport fields 24 positions, and for this reason, teams have much autonomy in how they choose to compose their roster from a positional standpoint. However, that being said, teams will generally comprise their team with; two quarterbacks, four

running backs and fullbacks, six wide receivers, three tight ends, nine offensive linemen, eight defensive linemen, seven linebackers, seven defensive, one punter, one kicker, and one long snapper (often listed as an extra tight end).

NFL Salary Cap

The NFL is what is known as a salary cap league. The salary cap can be defined as a set amount of money each year that teams may spend on player salaries to field their rosters. Practice rosters and salaries exist outside the salary cap, with players getting weekly salaries based on experience and ability. Players with two or fewer years of experience get paid \$12,000 per week, and veteran players with more than two years of experience get paid \$16,000 - \$20,600. While many professional sports leagues operate within a salary cap system, the NFL is more stringent with its cap rules because they are a "hard cap" league. The difference between a "hard cap" as opposed to a "soft cap" revolves around specific rules that may allow a team to spend in excess of the salary cap value. For example, in leagues such as the NBA, teams hold certain rights towards tenured players, allowing them to sign player contracts that exceed the established cap. The NFL, however, has no exceptions to exceeding the cap, and every participating team must stay under the cap with no exceptions. The current NFL salary cap is determined as a percentage of the revenues generated by the NFL. Under the current CBA (collective bargaining agreement between players and owners), players are entitled to 48% of the league's revenues in the form of the salary cap. Teams can also carry over past years' cap savings to add to the present year's cap total. Teams must also spend above a salary floor to players over the four-year CBA terms, and if this amount is not met, any money below the set floor will be distributed to players who made up the roster over the four-year CBA agreement. Currently, the

cap spending floor is set at 89%. However, the NFL as a league must spend 95% of the cap on average across all teams during the four-year CBA. The same penalties apply for the league for missing the 95% minimum, except salaries would then be distributed to all players on any active roster over the four years.

NFL Contract Construction

NFL contracts are composed of two primary types of money: the "base salary" and the "signing bonus." The signing bonus is guaranteed (guaranteed to be entitled to the player regardless of whether they are on the active roster or not) and given to the player upon signing their contract. "Cap hit", is the player's salary counted against the cap space; the signing bonus is prorated across the agreed-upon contract length so that the cap hit of the signing bonus is not attributed solely against one year's salary cap. The remaining contract amount not included in the signing bonus is spread across the contract length. For example, a contract of 10 million dollars over five years with a 5 million dollar signing bonus would work as follows: 10 million dollars in contract value, 5 million dollars paid up front to the player, 5 million dollars in total base salary, and a yearly cap hit of two million dollars which can be attributed to one million in base salary and one million in the guaranteed cap hit. The distinction of guaranteed money is significant in NFL contract discussions due to the high frequency of injury and players getting released to free agency. A player's base salary is not guaranteed to them; thus, in the advent of their release, they are entitled to none of the previously agreed upon compensation. For this reason, teams often will only prorate the guaranteed money to the first half of the contract, so in the advent of a dip in player performance in the later half of the contract, they may release the player without a cap penalty. Cap penalty or "dead money" is the money from a guaranteed

contract that now counts against the cap after a player has been released before the complete proration of the signing bonus. Depending on when a player is released (June 1 is the date as it marks the start of the new NFL contractual season), the remaining guaranteed money which has yet to be used against the cap will result in a "dead cap hit" totaling in all of the remaining guaranteed money hitting against the current year's cap, or this value being spread over two years (the current and ensuing season). Small sums of cap space will be assigned to player-based incentives. Stating that if a player, generally with statistical incentives, meets certain goals, they will be paid an extra sum of money. Incentives are categorized as "likely" and "unlikely," where likely incentives count against the cap at the start of the year and will be added to the following year's cap if not reached. Unlikely incentives are alternatively not counted against the cap at the beginning of the year and will subsequently be reduced from the following year's cap space if met.

NFL Rookie Contract Structure

As it currently stands, rookie contracts are four years long and have preset values established for base salary and signing bonus. Players drafted in the first round have fully guaranteed contracts (the signing bonus and base salary are guaranteed), and first-round rookies may be extended for a fifth year. In contrast, players drafted in any other round cannot have their original rookie contract extended for a fifth season and must alternatively sign their second contract for their fifth year of play. The only negotiating power players have is the payout terms of their guaranteed money and how much of that salary will come as a signing bonus. This motivation stems from players seeking their contract money sooner rather than later, so it serves them to negotiate for a more significant signing bonus (of the pre-established guaranteed

money). Players may only begin renegotiating their contract and contract extension talks after the conclusion of their third season in the NFL. This allows players on their rookie contracts to sign their second contract before their fourth season begins, so they are not playing on an expiring contract. The fear of playing on an expiring contract is a lack of security due to injury risk.

Significant Rule Changes

Football at the NFL level is currently played and is officiated far differently than it originally was 30 years ago in 1994. In addition to changes in officiating, the NFL has altered how they handle rookie contract negotiations. As mentioned earlier, both changes have directly affected the outlook for the NFL salary cap roster building. Directly in terms of rookie contract structure alterations and indirectly with rule changes altering the style of play and, in turn, affecting positional value. In 2011, the NFL altered rookie contract negotiation rules; prior to 2011, rookies were able to negotiate contract terms and total salary. However, after the 2011 CBA agreement, every rookie contract was preset and allowed no room for negotiation past how much of the guaranteed salary would be paid as a signing bonus. Rookie contract lengths were also all set at four years, with the option of a 5th-year extension for first-round selections. This rule change to the rookie contract structure is primarily attributed to the rookie contract signed by 2010 first-overall pick Sam Bradford. In 2010, Sam Bradford negotiated and signed a contract for seven years, 78 million dollars. Not only was this the largest rookie contract in history at the time, but it was so much money allocated toward a rookie player that the following year, the rule to cost control rookie contracts was implemented. For perspective, the 2023 first overall pick Bryce Young signed a 4-year contract, a \$37,955,071 contract. Regarding annual salary and total

contract value, Sam Bradford made more 13 years ago than current players picked in the same draft spot.

Significant changes to the game's rules were implemented in a series of player safety rules over the past fifteen years. The main goals of these rules were to protect the quarterback position and prevent blows to the head for all participating players. Most notably, a 2009 rule change preventing tackling quarterbacks inside the pocket (the pocket being the space formed on passing plays in between the offensive tackles) below the knee is banned. This followed an ACL injury to Tom Brady and was especially important as it was a rule that changed officiating to favor and/or value the quarterback above other positions. Between 2010 and 2013, strict rules preventing players from using their helmets to initiate contact, namely protecting "defenseless receivers" from being impacted in the chest up, are believed to open up the passing game. Another meaningful officiating change in 2011 emphasized illegal contact being officiated more strictly. Illegal contact is a five-yard penalty when a defender contacts a receiver attempting to evade him beyond five yards from the line of scrimmage. From 2010 to 2011, there was an over 90 penalty increase in defensive pass interference, holding, and illegal contact, increasing from 362 to 455.

Salary Cap - How it Changes

The NFL salary cap has not increased at a linear rate, and the changes in the cap are important because they can be compared with the increase in positional spending. When viewed over time, it can be determined that some positions are changing value at a different rate than others, depending on their deviation from the change in salary cap. Table 1⁴ shows the past ten

⁴ A table of the data used in figure 2. It is again a sample size of every rostered NFL player's salary by year, averaged by position group since 2000.

years of NFL salaries by position group, and Table 2⁵ compares the yearly change in positional salary expressed as a percentage. Regarding total salaries by position group, running backs have been the lowest-paid group every year of the past decade. Running Backs in the past decade have only twice outgrown the salary cap; this is in contrast to every other offensive position group, which has outpaced the salary cap four times, seven times in the case of wide receivers. However, this only tells part of the story of positional value. Within football, there are multiple players on the field at a time for every position except for running back and quarterback, where there is only one from the position group on the field at a time. For this reason, comparing numbers totals for a position group as a whole is not helpful to us, and relative change is better for the research conducted.

Table 1

Year	QB	RB	WR	TE	OL	IDL	EDGE	LB	S	CB	Cap Total
2023	\$16,556,878.72	\$8,408,472.88	\$22,638,406.88	\$9,543,803.16	\$37,124,798.22	\$20,282,891.72	\$20,435,958.25	\$12,662,393.97	\$13,088,229.22	\$16,777,265.78	\$224,800,000.00
2022	\$17,719,395.69	\$7,856,034.31	\$19,805,280.69	\$8,611,587.63	\$32,391,070.31	\$19,035,834.50	\$19,485,904.91	\$11,744,193.72	\$12,254,971.81	\$17,265,571.78	\$208,200,000.00
2021	\$15,949,603.19	\$7,648,576.03	\$17,932,450.31	\$8,451,436.75	\$30,730,010.25	\$17,218,043.19	\$18,050,949.03	\$12,574,640.16	\$11,229,305.22	\$15,863,669.56	\$182,500,000.00
2020	\$18,721,531.66	\$7,920,073.94	\$19,825,744.72	\$9,082,390.84	\$33,821,889.50	\$18,814,519.28	\$20,356,461.09	\$13,269,230.59	\$11,917,007.31	\$17,191,005.69	\$198,200,000.00
2019	\$17,975,651.69	\$6,668,023.00	\$18,765,885.19	\$7,784,424.06	\$32,367,404.91	\$15,784,935.00	\$17,391,187.88	\$12,604,315.41	\$11,340,261.75	\$17,343,206.84	\$188,200,000.00
2018	\$18,333,984.63	\$6,338,196.81	\$17,549,598.78	\$7,932,436.06	\$29,163,176.22	\$15,478,432.06	\$17,588,885.03	\$12,272,106.97	\$10,673,692.50	\$16,161,980.63	\$177,200,000.00
2017	\$16,169,610.25	\$6,856,225.84	\$16,499,023.19	\$8,025,190.09	\$26,451,020.66	\$15,130,327.41	\$16,767,474.78	\$10,736,801.41	\$10,359,074.88	\$14,719,157.41	\$167,000,000.00
2016	\$16,411,682.31	\$6,040,390.44	\$15,442,035.31	\$6,902,786.13	\$22,867,240.25	\$13,328,109.34	\$15,497,782.72	\$10,491,635.50	\$9,801,162.31	\$14,374,202.50	\$155,270,000.00
2015	\$12,873,642.28	\$5,981,942.16	\$13,583,709.06	\$6,816,372.13	\$21,168,644.47	\$12,639,522.63	\$14,833,015.22	\$10,249,771.38	\$8,879,123.78	\$13,115,046.03	\$143,280,000.00
2014	\$11,842,018.94	\$6,408,846.66	\$12,479,146.38	\$5,621,019.25	\$20,713,455.97	\$12,661,855.44	\$13,840,725.56	\$10,060,211.22	\$7,670,876.78	\$11,227,169.28	\$133,000,000.00

⁵ The same group is shown in Table 1 but expressed in terms of contract % change rather than dollar change by position.

Table 2

Year	ΔQB	ΔRB	ΔWR	ΔTE	ΔOL	ΔIDL	ΔEdge	ΔLB	ΔS	ΔCB	ΔCap Total
2023	-6.56%	7.03%	14.30%	10.83%	14.61%	6.55%	4.88%	7.82%	6.80%	-2.83%	7.97%
2022	11.10%	2.71%	10.44%	1.89%	5.41%	10.56%	7.95%	-6.60%	9.13%	8.84%	14.08%
2021	-14.81%	-3.43%	-9.55%	-6.95%	-9.14%	-8.49%	-11.33%	-5.23%	-5.77%	-7.72%	-7.92%
2020	4.15%	18.78%	5.65%	16.67%	4.49%	19.19%	17.05%	5.28%	5.09%	-0.88%	5.31%
2019	-1.95%	5.20%	6.93%	-1.87%	10.99%	1.98%	-1.12%	2.71%	6.24%	7.31%	6.21%
2018	13.39%	-7.56%	6.37%	-1.16%	10.25%	2.30%	4.90%	14.30%	3.04%	9.80%	6.11%
2017	-1.47%	13.51%	6.84%	16.26%	15.67%	13.52%	8.19%	2.34%	5.69%	2.40%	7.55%
2016	27.48%	0.98%	13.68%	1.27%	8.02%	5.45%	4.48%	2.36%	10.38%	9.60%	8.37%
2015	8.71%	-6.66%	8.85%	21.27%	2.20%	-0.18%	7.17%	1.88%	15.75%	16.82%	7.73%
2014	17.30%	-5.96%	9.05%	2.33%	7.92%	14.15%	11.74%	13.53%	3.64%	6.40%	7.61%

Empirical Analysis

The analysis conducted within this paper used a multitude of series and graphs, which work together to prove that a 2011 CBA rookie contract structure change is the main cog in running back devaluation. The charts and data will also work to disprove and contest any pre-existing notions that look to explain runningback devaluation in a mostly/purely non-contractual sense. Again, while some explanations others have made on the surface could explain running back devaluation, we argue these explanations were only made possible by the 2011 change. All statistical data for player production was taken from the *profootballreference* database, and all contract information was taken from *Sportrac*⁶, a sports contract history database.

League Averages

In attempting to solve the question at hand regarding offensive output, we do not see the change that would be expected given the most common preconceived notions. It has been a

⁶ Five contracts amongst the 480 plus taken were taken from online articles detailing the contract due to specific player contracts not being available on the database

running idea for the past decade or so that the NFL has grown to be a much more high-scoring, fast-paced game. However, when viewing the league average data of offensive plays and drives per game, we see this not to be the case entirely. As seen in Table 3⁷, the average points per team has essentially remained the same, with less than a 3-point increase in the past 30 years. Scoring averages for teams over the last three years have remained the same by one field goal on average. Going even further, there has been no significant change in average offensive plays per game, which rules out the possibility of more plays and higher scoring, driving some change in perceived positional value.

Returning to Table 3, we see an interesting shift in turnover frequency. In 1994, teams averaged 1.9 turnovers per game, and this value is now down 31.5% to 1.3 turnovers per game. This suggests that offenses have gotten more efficient while not seeing a massive surge in production or total plays. Table 4⁸ shows us that interceptions per game have decreased by 27% in the past 30 years, and Figure 3⁹ shows a surge in completion percentage from 58% to 65% in that same timeframe. While this does tell us that the league has become more efficient in their ability to throw the ball, there still has not been a significant jump in passing yardage output, attempts, or touchdowns. When considering table 5¹⁰, which shows league-wide averages in rushing; there is again another jump in efficiency. Fumbles are down 25% from 1994, and yards per rushing attempt have increased 13.5%. Again, this is done, however, with little to no movement in rushing yardage and touchdown outputs. It could be that there has been some philosophical change in how the offense is approached from a coaching perspective, or maybe

⁷ Table 3 shows the average offensive statistics of every NFL team by season since 1994, with the sample being the 32 NFL teams' statistics each year.

⁸ Table 4 shows the average passing statistics of every NFL team by season since 1994, with the sample being the 32 NFL teams' passing statistics each year.

⁹ Figure 3 shows the average completion percentage of every NFL team by season since 1994. The sample is the 32 NFL teams' passing statistics each year. The X-axis represents the season/year, and the Y-axis represents the NFL's average completion percentage.

¹⁰ Table 5 shows every NFL team's average rushing statistics by season since 1994. The sample is the 32 NFL teams passing statistics each year.

even players have gotten better. However, there is still nothing to suggest that from a league-wide perspective, one can assert that there has been a passing renaissance throughout the NFL, which is why the running back position is now non-valuable. There has been no statistical change in total offensive production, and the visible changes in passing (efficiency stats) can also be seen similarly in rushing. It can be asserted through the present data that changes in running back value can not and should not be attributed to "changes in how the game is played" or a "pass-happy league."

Table 3

Year	Tms	Points For	Yds	OffensivePlays	Y/P	TO	1stD	#Dr	% of Scoring Drives	% of Turnover Drives	Plays Per Drive	Yards Per Drive	Points Per Drive
2023	32	21.8	331.6	63.1	5.3	1.3	19.3	11.1	35.5	11.6	5.88	29.9	1.88
2022	32	21.9	340.1	63	5.4	1.3	19.7	10.9	36.2	11.3	5.95	31.1	1.92
2021	32	23	343.6	63.3	5.4	1.3	20.3	10.8	37.8	11.5	6.04	31.7	2.05
2020	32	24.8	359	64.3	5.6	1.3	21.7	10.9	39.8	11.4	6.09	32.9	2.2
2019	32	22.8	347.9	63.5	5.5	1.4	20.3	11.2	35.7	11.8	5.85	31	1.94
2018	32	23.3	352.2	62.9	5.6	1.4	20.4	11.2	36.5	11.8	5.8	31.4	2
2017	32	21.7	334.1	63.4	5.3	1.4	19.3	11.5	33.8	11.4	5.69	28.9	1.78
2016	32	22.8	350.4	63.9	5.5	1.4	20.3	11.4	35.6	11.3	5.77	30.6	1.91
2015	32	22.8	352.7	64.4	5.5	1.5	20.1	11.7	34.3	11.7	5.69	30.2	1.84
2014	32	22.6	348.1	64	5.4	1.5	20	11.5	34.1	12.1	5.71	30.1	1.83
2013	32	23.4	348.5	65	5.4	1.6	19.9	12	33.8	12.5	5.58	29	1.81
2012	32	22.8	347.2	64.2	5.4	1.6	19.8	11.7	33.7	12.5	5.67	29.7	1.79
2011	32	22.2	346.8	63.6	5.5	1.6	19.5	11.8	32.8	12.8	5.55	29.3	1.74
2010	32	22	336	63.1	5.3	1.7	18.9	11.7	32.4	13.3	5.54	28.5	1.74
2009	32	21.5	335.1	62.9	5.3	1.7	18.7	11.6	31.8	13.7	5.56	28.7	1.72
2008	32	22	327.2	61.9	5.3	1.5	18.4	11.3	34	12.9	5.64	28.8	1.8
2007	32	21.7	325.2	62.8	5.2	1.8	18.6	11.6	32.1	14.5	5.58	28	1.71
2006	32	20.7	322.1	62.5	5.2	1.8	18.4	11.6	30.9	14.2	5.53	27.6	1.65
2005	32	20.6	315.9	62.5	5.1	1.8	18.4	11.7	31	13.8	5.5	26.9	1.65
2004	32	21.5	327.2	62.5	5.2	1.8	18.8	11.7	30.8	14.3	5.47	27.8	1.69
2003	32	20.8	318.3	62.7	5.1	1.8	18.3	11.9	30.4	14	5.45	26.8	1.63
2002	32	21.7	328.4	63.6	5.2	1.8	19	11.8	31.4	14.4	5.56	27.8	1.71
2001	31	20.2	317.6	62.6	5.1	1.9	18	11.9	29.5	14.5	5.44	26.6	1.56
2000	31	20.7	319.4	63	5.1	1.9	18.4	11.9	30.2	14.7	5.46	26.8	1.62
1999	31	20.8	318.8	63.6	5	1.9	18.1	12.3	29.2	14.6	5.34	25.9	1.55
1998	30	21.3	317.7	63.1	5	1.8	18.2	11.2	30.4	14.1	5.4	26.5	1.63
1997	30	20.7	314.9	63.8	4.9	1.8	18.3						
1996	30	20.4	316.4	63.9	5	1.9	18.7						
1995	30	21.5	328.9	64.5	5.1	1.9	19.3						
1994	28	20.3	317.9	63.7	5	1.9	18.6						

Table 4

Year ▼	Tms ▼	Cmp ▼	Att ▼	Cmp% ▼	Yds ▼	TD ▼	Int ▼	Y/A ▼	Y/C ▼	Sk ▼
2023	32	21.7	33.7	64.5	218.9	1.4	0.8	7	10.9	2.6
2022	32	21.4	33.3	64.2	218.5	1.4	0.8	7	10.9	2.4
2021	32	22.3	34.4	64.8	228.3	1.5	0.8	7.1	11	2.3
2020	32	23	35.2	65.2	240.2	1.7	0.8	7.2	11.1	2.2
2019	32	22.1	34.9	63.5	235	1.6	0.8	7.2	11.4	2.5
2018	32	22.4	34.5	64.9	237.8	1.7	0.8	7.4	11.4	2.5
2017	32	21.2	34.2	62.1	224.4	1.4	0.8	7	11.3	2.3
2016	32	22.5	35.7	63	241.5	1.5	0.8	7.2	11.4	2.2
2015	32	22.5	35.7	63	243.8	1.6	0.9	7.3	11.5	2.3
2014	32	21.9	34.9	62.6	236.8	1.6	0.9	7.2	11.5	2.4
2013	32	21.7	35.4	61.2	235.6	1.6	1	7.1	11.6	2.5
2012	32	21.2	34.7	60.9	231.3	1.5	0.9	7.1	11.6	2.3
2011	32	20.4	34	60.1	229.7	1.5	1	7.2	12	2.3
2010	32	20.5	33.7	60.8	221.5	1.5	1	7	11.5	2.2
2009	32	20.3	33.3	60.9	218.5	1.4	1	7	11.5	2.2
2008	32	19.7	32.3	61	211.3	1.3	0.9	6.9	11.4	2
2007	32	20.4	33.3	61.2	214.3	1.4	1	6.9	11.2	2.2
2006	32	19.1	32	59.8	204.8	1.3	1	6.9	11.5	2.3
2005	32	19.1	32.2	59.5	203.5	1.3	1	6.8	11.4	2.3
2004	32	19.1	31.9	59.8	210.5	1.4	1	7.1	11.8	2.3
2003	32	18.9	32.2	58.8	200.4	1.3	1.1	6.6	11.3	2.1
2002	32	20.1	33.8	59.6	212.2	1.4	1	6.7	11.3	2.3
2001	31	19.2	32.6	59	205.8	1.3	1.1	6.8	11.5	2.4
2000	31	19.1	32.9	58.2	206.9	1.3	1.1	6.7	11.6	2.5
1999	31	19.3	33.8	57.1	212.3	1.3	1.1	6.8	11.8	2.5
1998	30	18.3	32.3	56.6	205	1.4	1.1	6.8	12.1	2.5
1997	30	18.4	32.8	56.2	201.8	1.3	1	6.7	11.9	2.6
1996	30	19.2	33.3	57.6	207.4	1.3	1.1	6.7	11.6	2.3
1995	30	20.2	34.8	58.2	220.8	1.4	1.1	6.8	11.6	2.2
1994	28	19.5	33.6	58	213.6	1.3	1.1	6.8	11.7	2.1

Figure 3

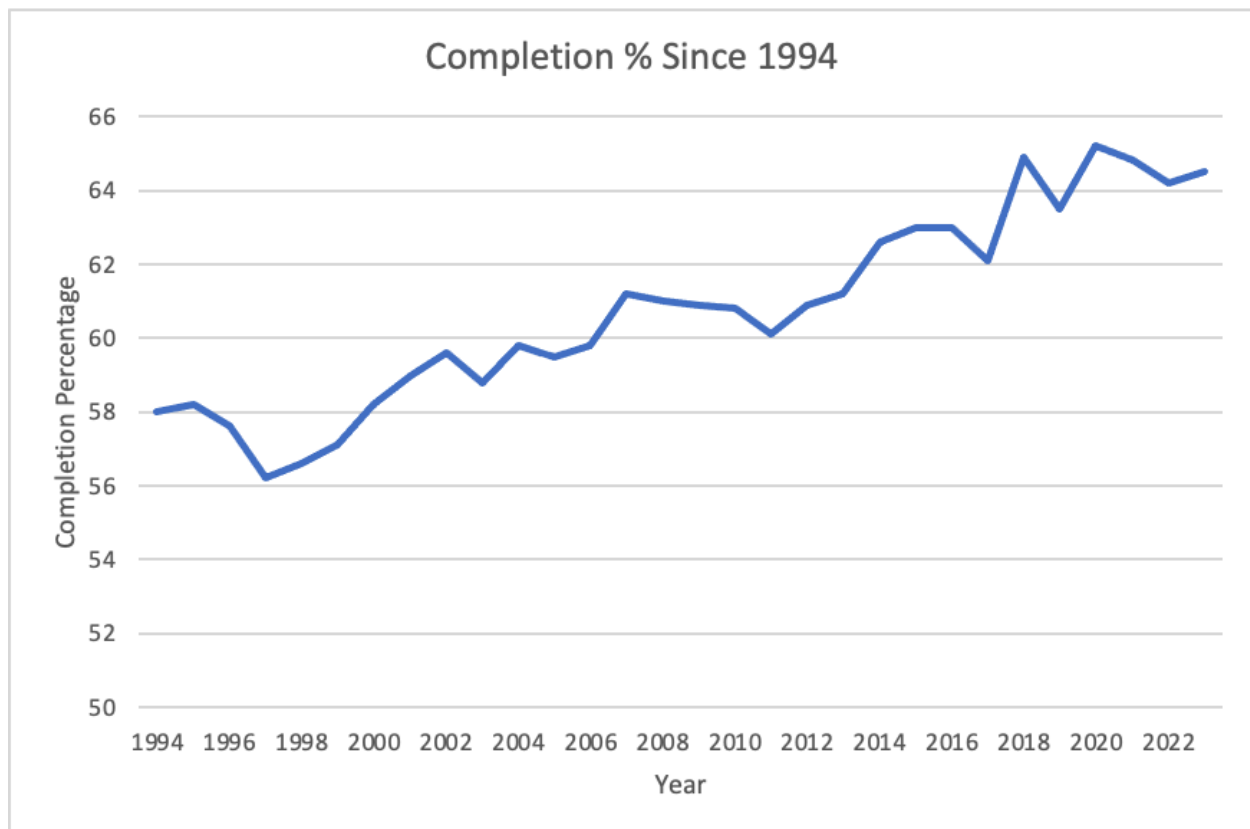


Table 5

Year	Tms	Att	Yds	TD	Y/A	Fmb	
2023	32	26.8	113	0.9	4.2	1.2	
2022	32	27.3	122	0.9	4.5	1.2	
2021	32	26.6	115	0.9	4.3	1.2	
2020	32	26.9	119	1	4.4	1.2	
2019	32	26.1	113	0.9	4.3	1.3	
2018	32	25.9	115	0.9	4.4	1.3	
2017	32	26.9	110	0.7	4.1	1.3	
2016	32	26	109	0.9	4.2	1.3	
2015	32	26.3	109	0.7	4.1	1.3	
2014	32	26.7	111	0.7	4.2	1.3	
2013	32	27.1	113	0.8	4.2	1.3	
2012	32	27.2	116	0.8	4.3	1.3	
2011	32	27.3	117	0.8	4.3	1.3	
2010	32	27.2	115	0.8	4.2	1.4	
2009	32	27.5	117	0.8	4.2	1.5	
2008	32	27.6	116	0.9	4.2	1.4	
2007	32	27.3	111	0.8	4.1	1.6	
2006	32	28.2	117	0.8	4.2	1.5	
2005	32	28.1	113	0.8	4	1.6	
2004	32	28.2	117	0.8	4.1	1.6	
2003	32	28.3	118	0.8	4.2	1.6	
2002	32	27.5	116	0.9	4.2	1.6	
2001	31	27.6	112	0.7	4.1	1.7	
2000	31	27.6	113	0.8	4.1	1.6	
1999	31	27.3	107	0.7	3.9	1.6	
1998	30	28.3	113	0.8	4	1.5	
1997	30	28.4	113	0.8	4	1.7	
1996	30	28.3	109	0.8	3.8	1.6	
1995	30	27.5	108	0.8	3.9	1.6	
1994	28	28	104	0.8	3.7	1.6	

Positional Contributions

With the onset of running quarterbacks such as Lamar Jackson and Jalen Hurts, as well as hybrid receivers like Deebo Samuels. Another emerging reason for running backs being devalued is that they are, in essence, being boxed out. As shown in Table 3, only so many yards are available for the players on each team to generate. So, we looked to answer if running backs are contributing less due to other positions' contributions to the run game and yardage generation in general. Figure 4¹¹ Shows the percentage of yards running backs contribute to the top 150

¹¹ The sample in Figure 4 is the NFL's 150 leaders in yards from scrimmage yearly since 1994. This sample includes the four offensive skill positions, which may possess the football, quarterback, running back, wide receiver, and tight end. (next page)

yardage generators, and Figure 5¹² it is the same group, but this time, the running back's share of touches. Yardage here means yards from scrimmage, a combination of both rushing and passing yards and touches refer to the combinations of rushing attempts and receptions. In both figures, a noticeable dip in the share running backs occupy of yards and touches occurs around our key date of 2011. This dip could also support the notion that running backs aren't as featured in the current NFL offensive landscape, but to make that assertion, we must also look at which positions now occupy the share of yards and touches that running backs have declined in.

Figures [6, 7, 8, & 9]¹³ These are the same charts as 4 and 5, except with the wide receiver and tight end positions, respectively. In Figures 6 & 7, there is a slight uptick in the amount of touches receivers compose of the group and a fluctuating dip in the receiver's share of yardage. Returning to table 4, this tracks with our league passing averages as a higher completion percentage without change in attempts or yardage implies more catches, with each catch being for fewer yards; table 4 shows this in the yards per completion column labeled "y/c." Most notably, in the tight end graphs in figures 8 and 9, there is a clear jump in the past 30 years in the share of yards and touches occupied by the tight end position. From this data, we can gather that running backs have experienced a sort of "boxing out" effect in terms of their positional contribution to yardage generation due to the jump in tight-end production. While the running back position has seen the most considerable negative change in their share of production, it is hardly to the degree that could warrant the complete stagnation of their contract growth. If a 5% drop in yardage share were to be used as the justification for running back

Yards from scrimmage can be defined as the combined rushing and receiving yards generated. The X-axis is the year/season, with the y-axis representing the percent of total scrimmage yards generated by the top 150 leaders that running backs account for.

¹² Figure 5 has the same group and x-axis as Figure 4; however, the y-axis now represents the percentage of touches amongst the top 150 NFL yardage leaders that are accounted for by the running back position. A touch can be defined as either a rush or a reception.

¹³ Figures 6 - 9 were created with the same sample group as Figures 4 & 5. However, the y-axis in each graph is now representative of the two new position groups, wide receiver and tight end. 6 and 7 represent the wide receiver's share of yards and touches, with 8 and 9 being the tight end's share of yards and touches. Again, charts 4-9 share a sample size and group; said group being the 150 scrimmage yards leaders by year in the NFL.

devaluation, then Figure 2 would not have shown wide receivers outpacing tight ends in contract growth this past decade, given it was the tight end position replacing the running backs decrease in shares; not the wide receivers.

This is not to disregard the drop in running back a share of yardage or touches but rather to make a note of an alternative explanation. All the yardage and touch share figures in this section see their most significant change around 2011. This is no coincidence, as 2011 was the year the NFL overhauled its rookie contract structure. This key date can explain all significant shifts in NFL statistics, both contractual and on-field production. Given the sudden shift in the running back and tight end share, it wouldn't make sense that a change in perceived positional value within the NFL outlook could be the reason for devaluation. This means that if teams truly were to value tight ends above running backs due to stylistic choices, this wouldn't suddenly manifest in a three-year window; it would be an incremental shift as the league's schematic philosophy adjusted. In addition, returning to tables 3, 4, and 5, the league has stayed the same in offensive numbers, so it wouldn't be logical for perceived positional value in terms of on-field contribution to change without a matching change in offensive output. Nevertheless, in order to explain the drop in running back contribution to the yardage leaders through the lens of a 2011 contract shift, we will first need to dig deeper into specific position production numbers.

Figure 4

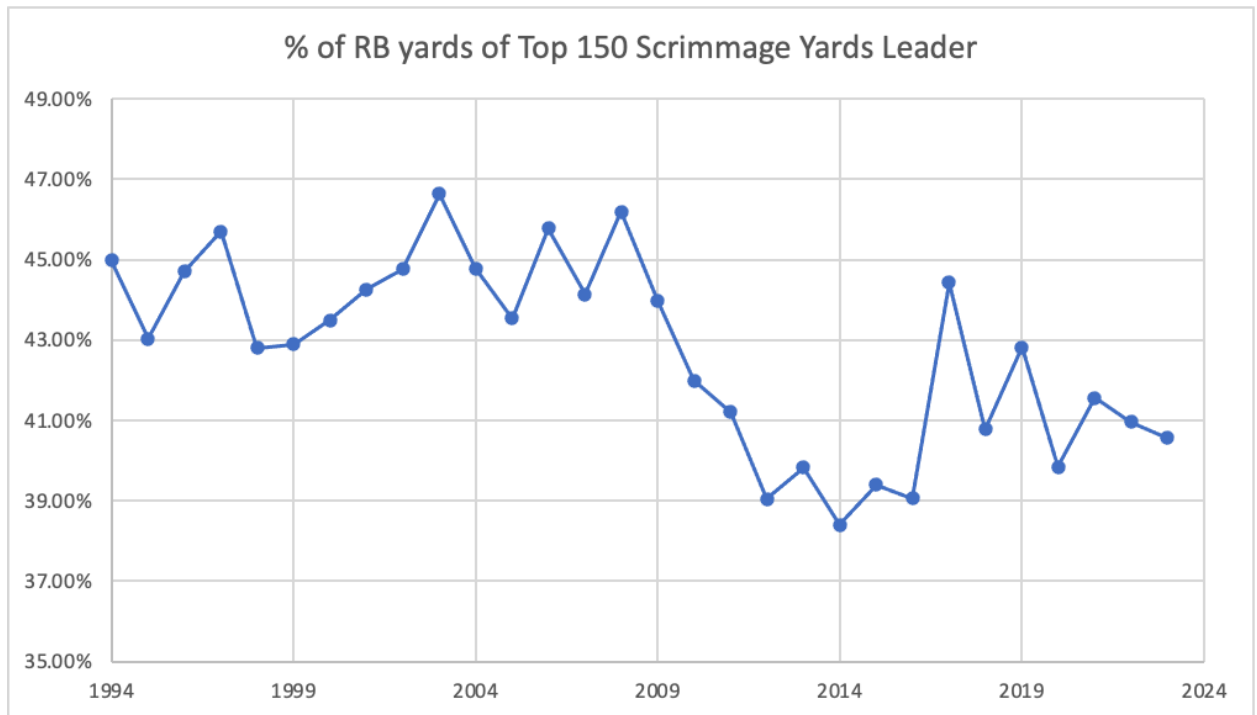


Figure 5

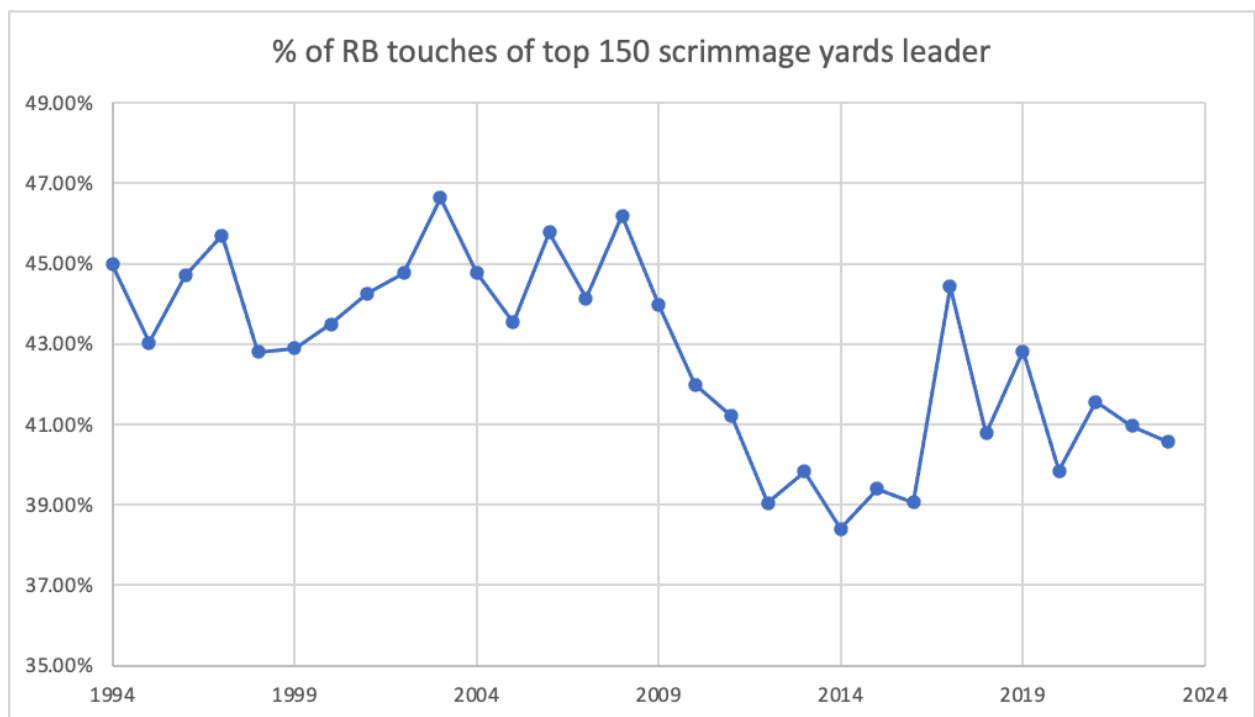


Figure 6

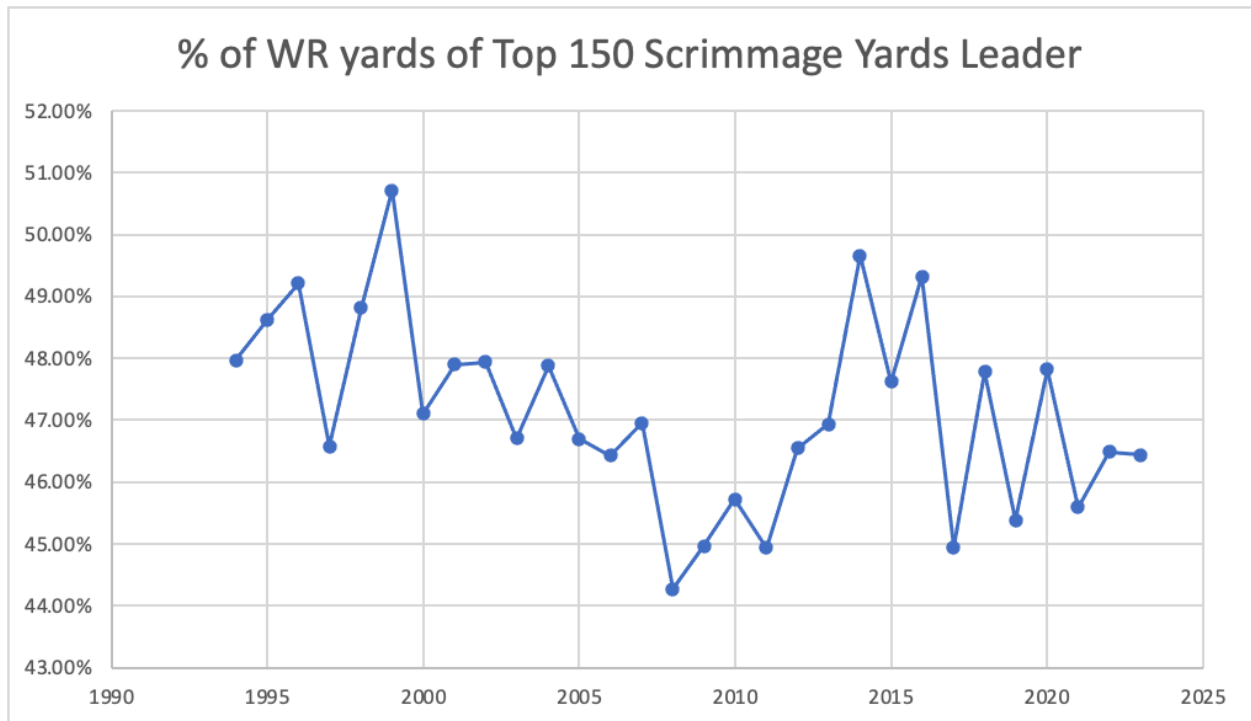


Figure 7

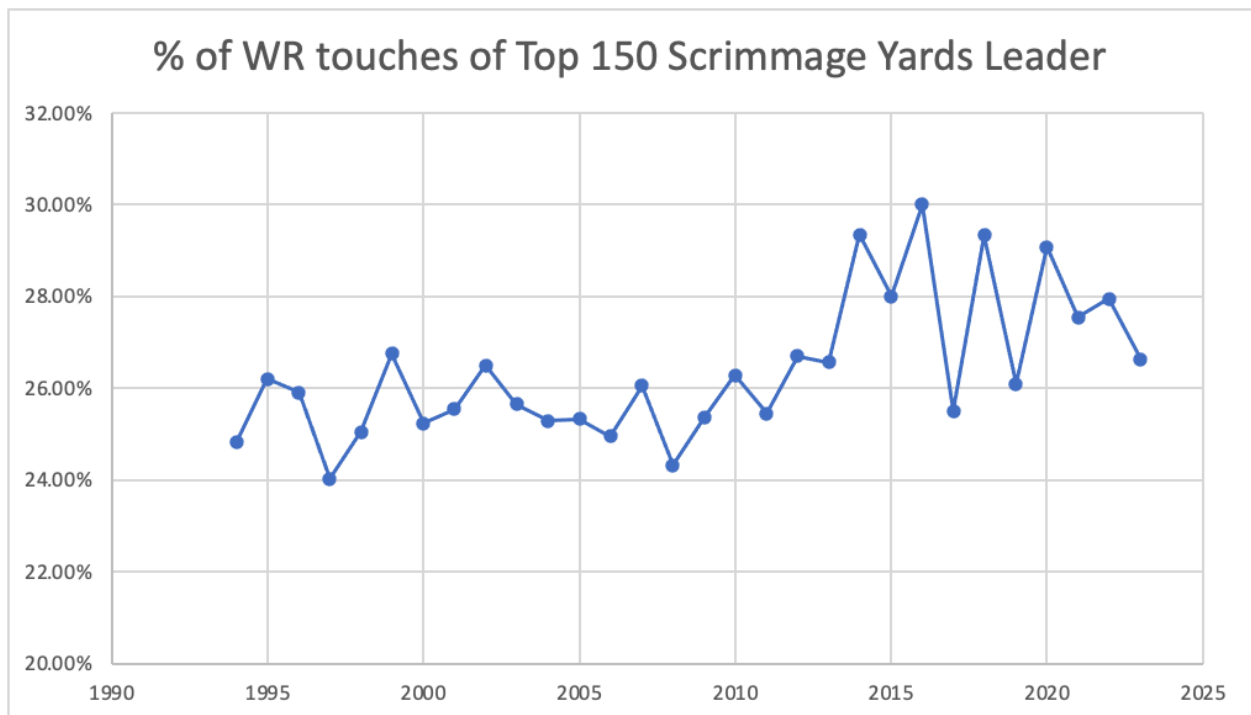


Figure 8

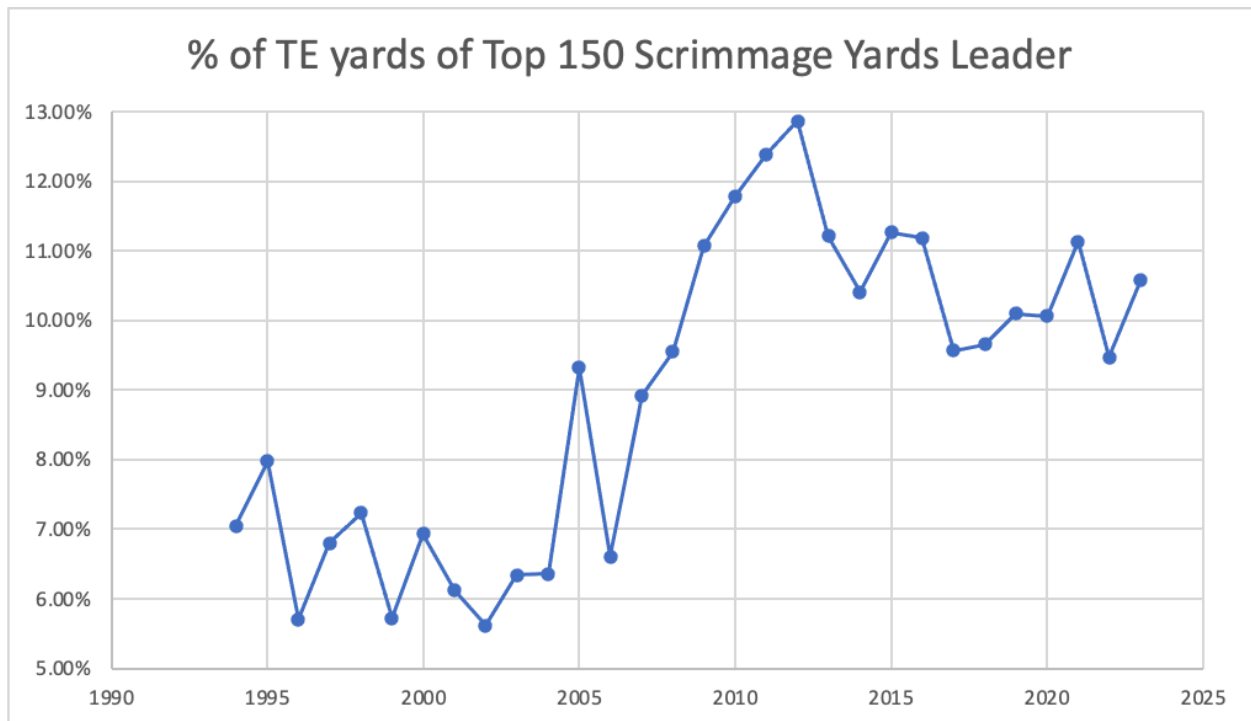
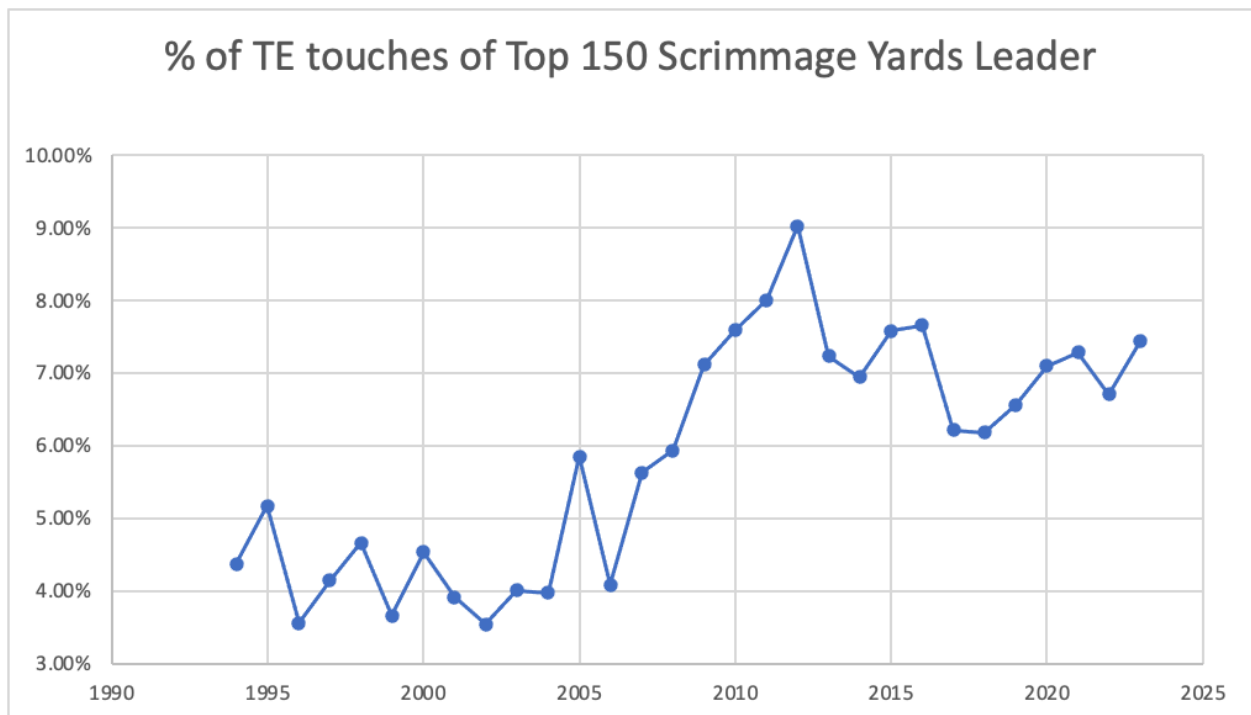


Figure 9



Output by Position

Running back by committee is the idea that the NFL rushing game can be formed around a group of running backs, sometimes including the quarterbacks, in opposition to the more classic lead back, who takes the majority of rushing attempts. Recall that Table 5 shows the average rushing yards per game has remained static for the past 30 years, so it can be ruled out that less rushing is occurring. Even in rushing attempts, this number has only decreased by about one carry per game, which is less than a 5% decrease in average attempts per game by team. Running backs by committee is a real change in roster construction, as when viewing Figure 10¹⁴, the average season yards of running backs who finish top 10 in rushing. The best running backs in the league are not nearly as productive as they once were; this is on top of an additional game being played for the past three seasons. A significant drop in top rusher performance over the past 20 seasons, paired with a static in rushing per game number can only be explained through an increase in the split of rushing yards between players on a team. Even when factoring top running backs in terms of rushing and receiving, we still find a significant drop in more recent years; this can be seen in Table 6¹⁵. We continue to find significant change at this 2011 contract structure change date. This is where we can explain the 5% drop in running back's share of top yardage generators and touches and the current decrease in top rusher production. Looking back to Figure 1, we see the splitting-off point of the runningback contract change to hit only on the same timeline as all of our production changes. The changes to the rookie contract structure have made the strategy of running back by committee team building much more possible. Recall that wide receivers are already in a "committee" position in that they already share a workload with

¹⁴ Figure 10 shows the running backs who have finished in the top ten in league rushing yards by season since 2000. The y-axis represents the yearly average of the top rushers, and the x-axis represents the year/season.

¹⁵ Table 15 has a sample of the top ten running backs in terms of scrimmage yardage by season since 2000. It displays the scrimmage averages of these backs by year and their average touches by year.

other receivers. This only exacerbates the 2011 contract change issue in that, as we see in Figure 11¹⁶. Top receivers have not had a major change in per-game production, yet their position group has grown the most in salary this past decade (see Figure 2). Rookie contracts are mandatory for four years at minimum length, with no option for renegotiation until after three completed seasons. This has narrowed the research question to how contract structure within the bounds of rookie contracts and the NFL salary cap devalued the position. Historically, a very productive player early into their rookie contract would look for immediate contract renegotiation, whereas this is currently against NFL bylaws. The consensus idea that running back by committee is the driver in running back devaluation is founded in truth; however, it is, in actuality, a symptom of a deeper contract issue. It is a contract issue that directly works against the historical lifespan of running backs and in favor of less physically demanding positions.

¹⁶ Figure 11 has a sample group of the yearly receivers who place in the top ten in receiving and their per-game averages. The per-game average was necessary to accommodate the extra-game players in the past three seasons. The x-axis is a season, with the y-axis being the per-game receiving averages of these top wide receivers.

Figure 10

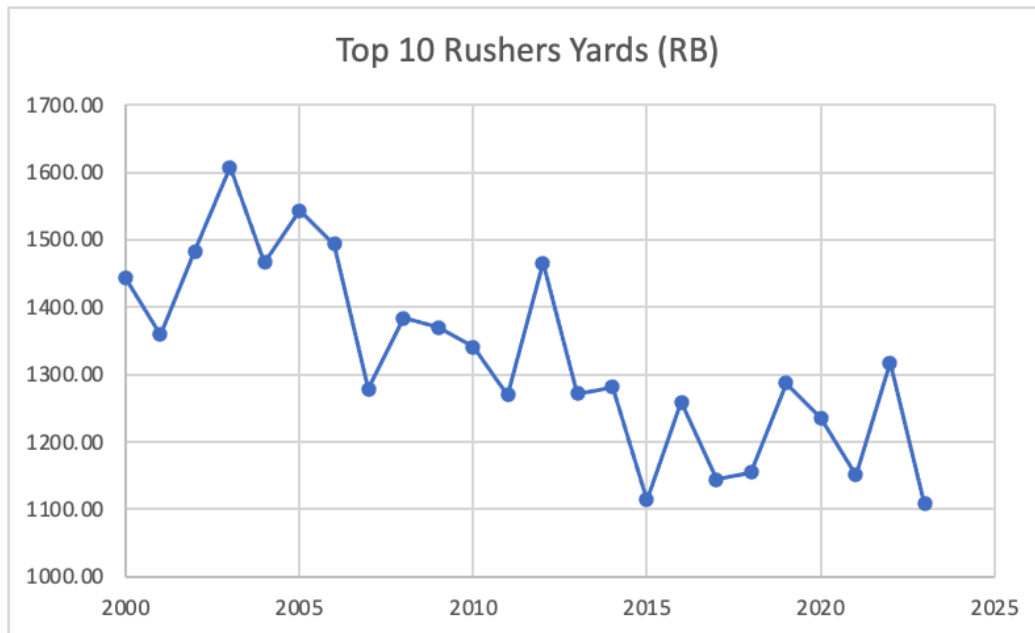
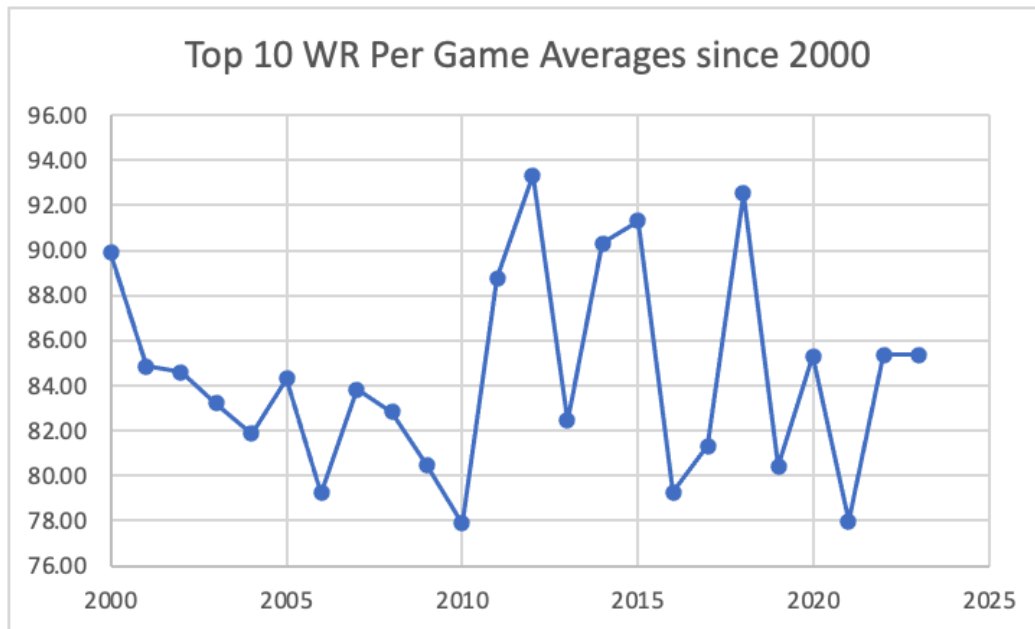


Table 6

Year	Avg Touches (RB Top 10)	Avg ScrYds (RB Top 10)
2023	303.60	1511.80
2022	320.50	1680.60
2021	301.40	1495.40
2020	293.70	1543.30
2019	315.00	1706.00
2018	299.90	1642.60
2017	312.20	1596.40
2016	308.40	1664.20
2015	287.20	1414.00
2014	321.50	1685.00
2013	318.10	1651.70
2012	331.70	1734.90
2011	318.40	1641.30
2010	324.00	1744.10
2009	327.90	1715.80
2008	338.50	1630.40
2007	334.40	1618.30
2006	374.70	1933.10
2005	370.00	1822.10
2004	372.40	1788.20
2003	392.30	2013.50
2002	364.70	1941.20
2001	365.90	1788.30
2000	357.60	1879.80

Figure 11



Positional Longevity

Age and overall longevity have often been cited as the primary reason for running back salary growth stagnation. Figure 12¹⁷ and table 7¹⁸ shows how the age of running backs who finish in the top 100 in rushing has trended since the year 2000. The downward trend of Figure 12 began before our key 2011 date. However, we see the most dramatic drop in age after 2011. Nevertheless, this 20-year downward trend could lead one to give credence to the notion that age is the cause of devaluation, thus a movement toward younger running backs. Although, figure 13¹⁹ and table 8²⁰ shows the same information as Figure 12 and Table 7 but with receivers and receiving yards, which gives us insight that receivers are getting younger as well. Receivers have

¹⁷ Figure 12's sample group consists of the running backs who have finished in the top 100 in league rushing totals by season since 2000. The x-axis shows the year/season, and the y-axis shows the average age of these running backs.

¹⁸ Table 7 contains the same sample group and information but is presented as a table so that specific yearly values in average top 100 running back age can be viewed.

¹⁹ Figure 13's sample group consists of the wide receivers who have finished in the top 100 in the league, receiving totals by season since 2000. The x-axis shows the year/season, and the y-axis shows the average age of these wide receivers.

²⁰ Table 8 contains the same sample group and information but is presented as a table so that specific yearly values in average top 100 wide receiver age can be viewed.

not been subject to longevity critiques, yet they, too, are getting younger as a position group. While still remaining older on average as a position group, receivers have had a larger shift down in average age than running backs. When viewing the age of positional leaders in terms of top ten rushers and receivers in figure 14²¹ and table 9²² we still see this downward trend in age for both position groups, even at the highest performance levels. Both position groups show trends in the direction of becoming younger but in contrast to the contract trend. This raises a contradiction to age being the driver of running back devaluation. We do not disagree with the notion that "running backs don't last"; our data even supports this in that productive running backs are noticeably younger than productive wide receivers. However, the short shelf life of running backs is not a new concept, and historical durability concerns would not make the position presently trend younger, especially as rules have been implemented to make the game safer. This is to say, it's understood that running backs do not last long, but they never have, so what is driving not only their but also receivers' average age down? The peak age of both groups is decreasing, but if only one is facing contract stagnation, then there must be a reason beyond literal age driving the change.

Going back to our group of the 150 players by year who have led the NFL in yards from scrimmage, we can view the change in running back age here to see if receiving or specialization protects backs from this drop in an early age peak. Figure 15²³ It is more of the same in that even when weighing receiving stats; this does not change the early peak of the running back. Again, we look to tie back this downward trend of both positional ages to this 2011 change in the

²¹ In Figure 14, we combine the groups of Figures 10 and 11, taking the running backs who finish in the top 10 in rushing by year and the receivers who finish in the top 10 in receiving by year. We used a y-axis of the average age of these two groups and an x-axis of the year/season.

²² Table 9 takes the same sample groups as Figure 14 but represents these values as the average ages of the position leaders by year in a table format. This is done so that specific age values may be referred to.

²³ We return to the earlier sample group of the top 150 scrimmage yards leaders—the graph sorts by an x-axis of year/season, with the y-axis reflecting average age. We then charted the average age of running backs who made the top 150 scrimmage leaders by season.

contract structure but also explain why receivers are still being paid more than RBs despite also getting younger. While it may seem as though a 1-2 years difference is insignificant, for this research, it does make a world of difference. NFL rookies predominantly enter the league at 21-22 years of age. Since rookie contracts expire four years (sometimes five years in the case of first-round picks) after league entry, the average age of players' eligible for their second contract is 25-26. We see that running backs in more recent years have fallen just on or under this second contract age, whereas wide receivers have remained firmly above until these past couple of seasons. This builds evidence to the potential that it is not the actions or lack thereof of running backs that have led to their lack of both pay and relative growth in pay. Instead, the replacement value and age of "peaking" for the wide receiver position fall much better in line with the current NFL contract structures.

Long-term roster sustainability has also become possible under the 2011 contract changes. Teams could generate more value from being built around a passing game, which, in terms of player sustainability, is the more practical option due to quarterbacks and receivers more commonly remaining successful later in their careers. This idea of roster sustainability works well with the 2011 change in rookie contract structure because a team can pay their QBs and WRs to long-term deals while still having talented running backs play through the draft. All while being able to fit this under the salary cap constraints given rookies cannot negotiate market value contracts anymore, thus allowing the running back to be underpaid for their prime years. So, it is not that runningback longevity is not a factor in their devaluation; it is more so that using their age against them contractually has only become possible since 2011. Historically, drafted running backs not only had the right to negotiate market value contracts for their expected contracts, but veterans in turn, had leverage to stay with their current team long term because

replacing them wasn't necessarily cheaper, not to mention the expense of a draft pick and uncertainty of rookie players panning out in the league. However, the question is how receivers are shown to have longer, more successful careers than running backs, getting younger on average and still having the most growth in terms of salary? The NFL draft and its cost-controlled contracts have affected both receivers and running backs, but receivers still benefit from later peaks (relative to running backs), ease of replacement in the draft, and less intense production fall-offs.²⁴

Figure 12

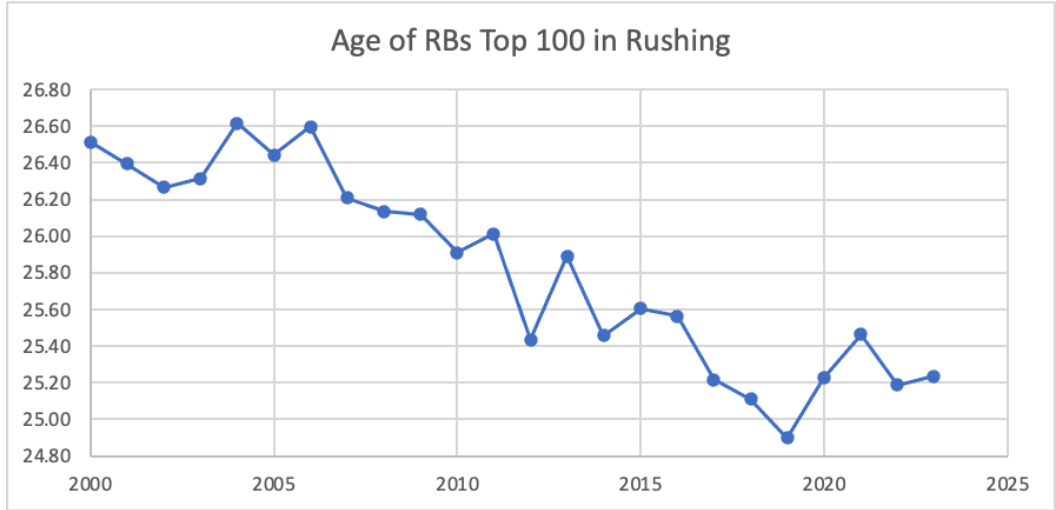


Table 7

²⁴ In 2023 alone, over ten receivers ages 30 and above finished in the top 100 in total receiving yards, and only two running backs over 30 were in the top 100 rushers.

Year ▼	Top 100 Rushers Age (RB) ▼
2023	25.24
2022	25.19
2021	25.46
2020	25.23
2019	24.90
2018	25.11
2017	25.22
2016	25.56
2015	25.60
2014	25.46
2013	25.89
2012	25.44
2011	26.01
2010	25.91
2009	26.12
2008	26.14
2007	26.21
2006	26.60
2005	26.44
2004	26.62
2003	26.31
2002	26.27
2001	26.39
2000	26.51

Figure 13

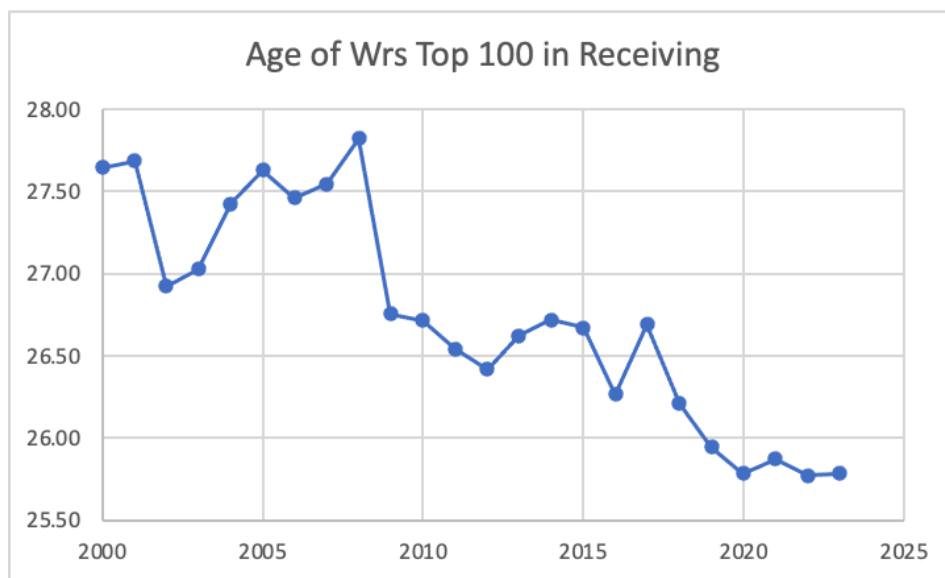


Table 8

Year	AvgWrAge
2023	25.79
2022	25.77
2021	25.87
2020	25.78
2019	25.94
2018	26.21
2017	26.69
2016	26.27
2015	26.67
2014	26.72
2013	26.62
2012	26.42
2011	26.54
2010	26.71
2009	26.76
2008	27.82
2007	27.55
2006	27.46
2005	27.63
2004	27.43
2003	27.03
2002	26.92
2001	27.68
2000	27.65

Figure 14

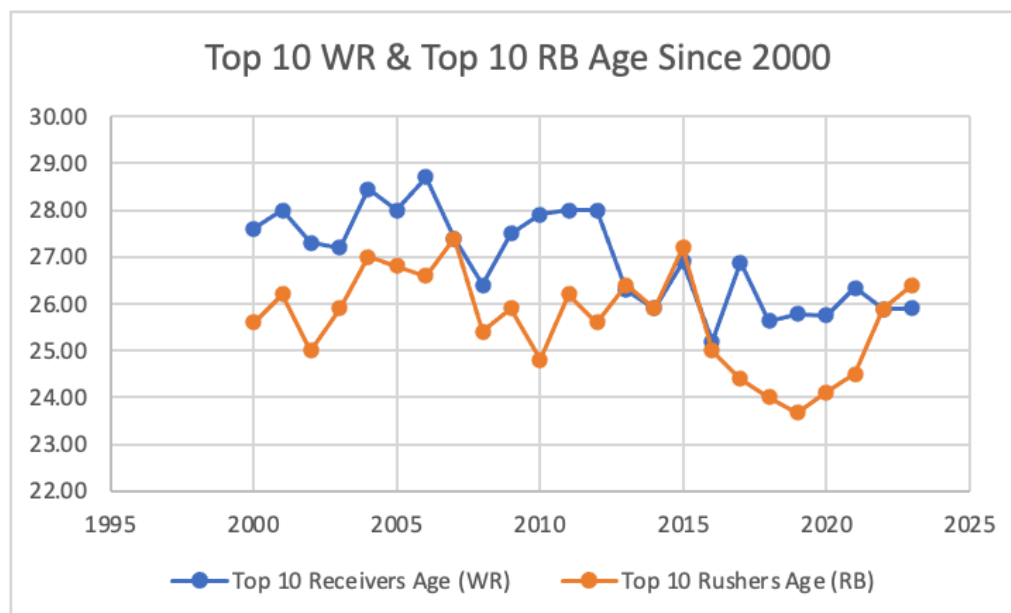
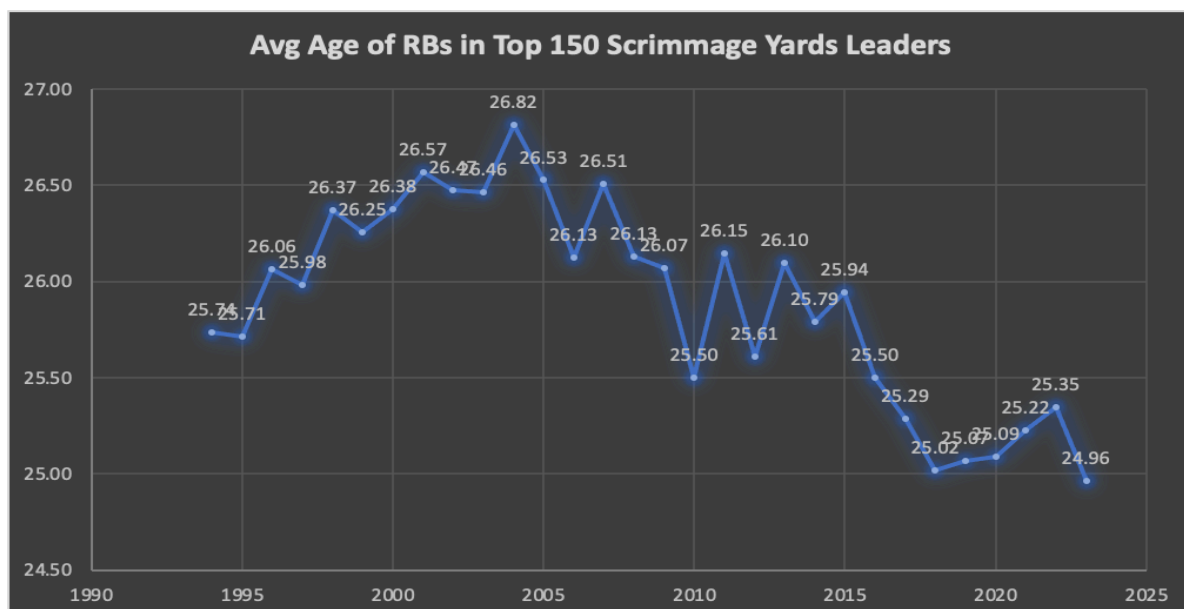


Table 9

Year	Top 10 Receivers Age (WR)	Top 10 Rushers Age (RB)
2023	25.90	26.40
2022	25.89	25.89
2021	26.33	24.50
2020	25.75	24.11
2019	25.78	23.67
2018	25.63	24.00
2017	26.89	24.40
2016	25.20	25.00
2015	26.90	27.20
2014	25.90	25.90
2013	26.30	26.40
2012	28.00	25.60
2011	28.00	26.20
2010	27.90	24.80
2009	27.50	25.90
2008	26.40	25.40
2007	27.40	27.40
2006	28.70	26.60
2005	28.00	26.80
2004	28.44	27.00
2003	27.20	25.90
2002	27.30	25.00
2001	28.00	26.20
2000	27.60	25.60

Figure 15



The NFL Draft

The premier rounds of the NFL draft are considered to be found in the first three rounds of the seven-round draft. Each team is given a single pick per round, which they will use to "draft" incoming NFL players, who almost always come from the NCAA college football circuit. Teams can also get extra draft selections in rounds 3 through 7 depending on the tender placed on a player who signs with or is traded to another franchise on the expiration of their rookie contract. Any draft pick held by a franchise can be traded to another franchise for almost always a player(s), a draft pick(s), and rarely a coach. Von Eikan (2017) found that using the variable CarAV (CarAV is an estimation of a player contribution found from their on-field statistics), provided by sports statistics site *profootballreference*, that through a sample size of NFL drafts from 1999 - 2008, a predictive assertion could be made on a player's contributed value based on draft selection. Unsurprisingly, he found that players drafted in rounds 1 were the most effective throughout their careers, with these contributions tapering downwards in a decline with the subsequent rounds. However, given the hard salary cap and the 2011 CBA setting rookie contracts as a non-negotiable pre-slotted value, the value of draft picks has primarily been altered in a post-2011 NFL. While it is true that first-round picks will most likely become the superior players, given their larger salary cap hits, they are not always the best generators of value relative to later-round players in terms of what franchising is paying. Kraeutler et al. (2018) also found that in almost all measurable statistics, first rounders outperformed players from all other rounds, which is again expected. However, when weighed against salary cap hit values, players selected in rounds 2-3 proved to be the most effective players in terms of the combined lens of production and contract.

Viewing the current NFL landscape in terms of trading draft capital picks. There has been much debate recently about the validity of drafting running backs using premier draft picks and that the devaluation of the position can be easily attributed to not using valuable draft capital on lower-value positions. Figure 16²⁵, table 10²⁶, and figure 17²⁷ shows how and where running backs have been drafted since the 2000 season. Our data displays yearly movement in the quantity and position of drafted running backs; the scale of the yearly fluctuation gives no strong indication that draft perceptions of running backs have changed significantly. Anecdotally, this past draft saw two running backs selected within the first half of the draft, the first half of round 1 being the most coveted picks. This is not necessarily to say that running backs are still being picked at their historical rate in the first round; one could claim that they are not when viewing the data. Rather, the NFL draft is such a year-by-year experience that teams will look to draft talented players with less emphasis on positional value, contrasting the free agency/extension market. The 2011 change in contract structure would not necessarily affect the draft position of running backs in that the rookie contracts themselves devalued the backs looking for their second contract, not the backs looking to be drafted. The four-year nature, along with preset salary numbers, essentially allows teams to lock players into underpaying deals, and by the time the running back is eligible to be extended, they can be replaced by a cheaper rookie back who can reach his peak faster than other positions such as WR (see table 7 & 8). Furthermore, any decrease in teams' propensity for drafting running backs in the first round does not come from first-round running backs being less successful; rather, it stems from the "lottery" nature of

²⁵ Figure 16 shows the yearly breakdown of running backs drafted by round. The x-axis shows the yearly draft class, and the y-axis shows the number of backs drafted. The colors subdivide the vertical axis into the number of running backs drafted by round.

²⁶ Table 10 is the same group as Figure 16, the yearly running backs drafted by round. It is formulated as a table for exact numbers.

²⁷ In Figure 17, the group is now the yearly running backs drafted in rounds 1 - 3. The x-axis represents the year, and the y-axis expresses the amount of backs drafted in rounds 1-3.

finding a talented running back in cheaper later-round contracts. This method can still be argued against, as Von Eikan (2017) found running backs to be amongst the most valuable first-round positions to be drafted, given that their "hit" rate is higher than that of other positions. Even moving into the current outlook of NFL positional value, a good running back is worth more than a first-round miss; thus, running backs still present a particular security in the first round, according to some schools of thought.

Figure 16

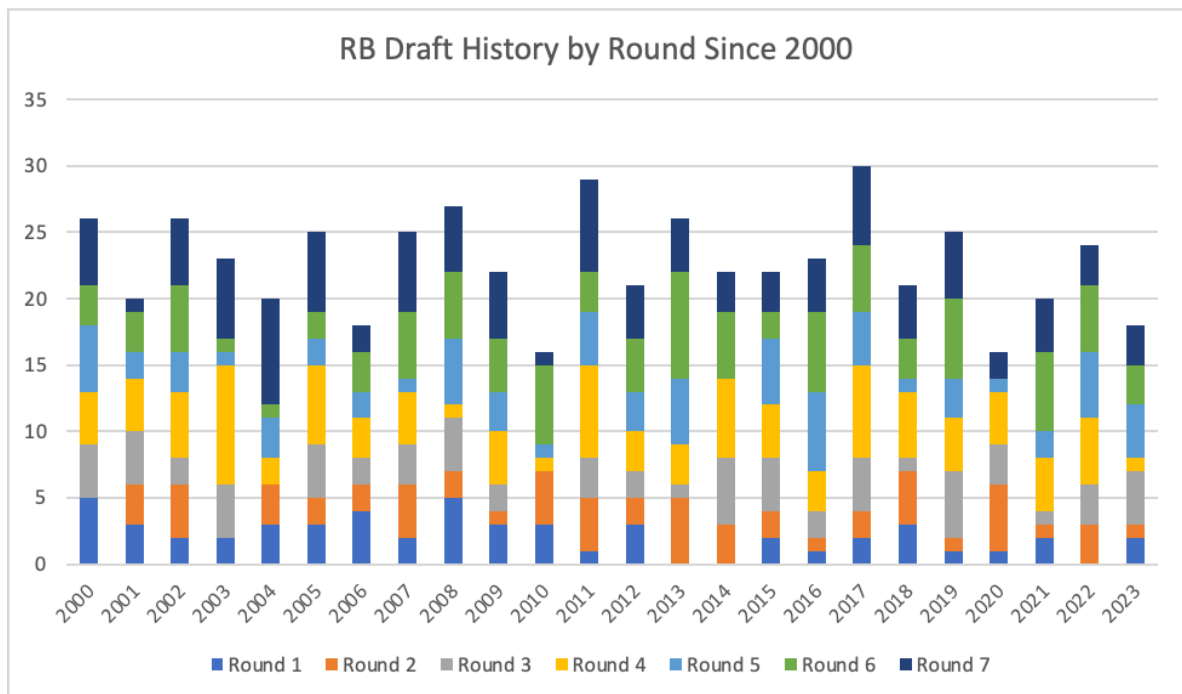
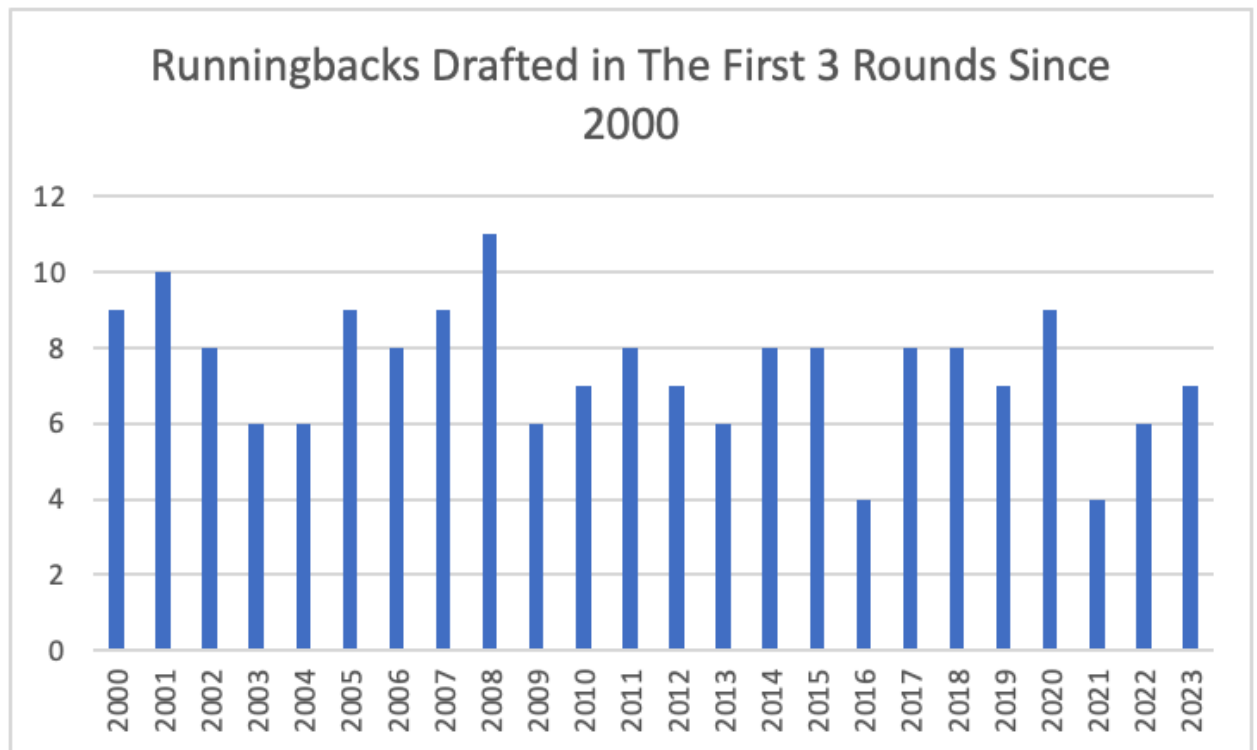


Table 10

Year	Round 1	Round 2	Round 3	Round 4	Round 5	Round 6	Round 7	Total
2023	2	1	4	1	4	3	3	18
2022	0	3	3	5	5	5	3	24
2021	2	1	1	4	2	6	4	20
2020	1	5	3	4	1	0	2	16
2019	1	1	5	4	3	6	5	25
2018	3	4	1	5	1	3	4	21
2017	2	2	4	7	4	5	6	30
2016	1	1	2	3	6	6	4	23
2015	2	2	4	4	5	2	3	22
2014	0	3	5	6	0	5	3	22
2013	0	5	1	3	5	8	4	26
2012	3	2	2	3	3	4	4	21
2011	1	4	3	7	4	3	7	29
2010	3	4	0	1	1	6	1	16
2009	3	1	2	4	3	4	5	22
2008	5	2	4	1	5	5	5	27
2007	2	4	3	4	1	5	6	25
2006	4	2	2	3	2	3	2	18
2005	3	2	4	6	2	2	6	25
2004	3	3	0	2	3	1	8	20
2003	2	0	4	9	1	1	6	23
2002	2	4	2	5	3	5	5	26
2001	3	3	4	4	2	3	1	20
2000	5	0	4	4	5	3	5	26

Figure 17



Current NFL Trades & Real Player Valuations

Chiefs & Titans Trade for All-Pro Cornerback

A recent trade between the Kansas Chiefs and Tennessee Titans saw all-pro corner L’Jarius Sneed trade a 2024 7th-round draft selection to the Titans in exchange for a 2025 3rd-round and a 2024 7th-round draft selection to the Chiefs. Sneed (27) is widely considered one of the NFL's premier cornerbacks, supported by his all-pro credentials from this past season. The Chiefs would be hard-pressed to replace a player of his talent in the first round, let alone the NFL draft. This begs the question as to why the Chiefs would accept a third-round pick as compensation for a player of this caliber. This can be explained through contracts and salary cap constraints. As of the writing of this thesis, the Kansas City Chiefs currently have \$25,410,425 in available cap space. When factoring in the cap space needed to replace existing players with players in the upcoming draft, this cap number could dip closer to \$20,000,000. After trading for L’Jarius Sneed, the Tennessee Titans signed him to a 4-year 76,000,000 contract, with a \$9,000,000 salary cap hit this upcoming 2024 season and a \$24,000,000 cap hit for the 2025 season. If the Chiefs were to attempt to sign Sneed to this same contract, they would have some breathing room for the 2024 season but would be pressed if not over the 2025 hard-cap amount. Exceeding the hard cap number would force dramatic roster alterations; an example is the Buffalo Bills, who have been forced to cut six-plus starters to meet cap constraints. For the Titans, the cost of acquiring Sneed was the third-round pick price tag and the massive contract extensions due to acquiring a player on an expiring deal. Even if the Chiefs were able to attain the Titan's 7th overall first-round selection in exchange for Sneed, the cap penalty of drafting a player that high in the first round would still be significant at a starting cap hit estimate of \$4,629,418 in the first year, growing to close to \$9,000,000 in year four. Alternatively, a 2025

3rd round selection will only hold an average cap hit of about \$1,000,000 annually. At the same time, the first-round selection is much more likely to be a successful player in the NFL than the third-rounder. The third-round selection gives the Chiefs a likely serviceable player who they maintain under contract for the next four years, and they can still be situated with good cap mobility, especially given the massive contracts of Chris Jones and Patrick Mahomes constraining them.

Talented Quarterbacks on Expiring Rookie Contracts

Justin Fields and Zach Wilson. Fields (25) and Wilson (24) were first-round quarterback selections of the 2021 NFL draft. Neither have had productive careers up to this point. However, the consensus remains that both remain immensely physically talented players, as reflected in their selections of 11th and 2nd overall. Neither have had any major injury nor aged enough into their career to consider these variables a factor. However, when these players shopped in the trade market, there were few buyers. Zach Wilson has yet to have any market in terms of interested franchises, and Fields was traded into a backup role for a sixth-round selection. It is almost impossible to find a player of either their talents in the sixth round or undrafted free agency, so why can they not find an interested team? Contract situations are the key for both of these players. Fields and Wilson, given that they are entering the last year of their rookie deal, are slated to become free agents after the 2024 season. The options for interested Teams looking for a developmental backup is to trade for these players and be forced to extend them for a deal that would certainly exceed \$1,500,000 given veteran minimums and backup status, not to mention both players likely not accepting a longer contract in hopes of proving their value to sign a better deal later. Or use the draft capital spent to acquire one of these players and draft a developmental rookie who will be paid half as much as these two quarterbacks and be able to

retain their much cheaper services for 4-years, given the 2011 CBA agreement on rookie contracts. It's not that these quarterbacks are devoid of the ability to contribute to teams; more so, they do not maintain the same financing options or benefits that a rookie player does.

Joe Mixon vs Christian McCaffrey

Joe Mixon (27), a pro-bowl running back for the Houston Texans, was traded to the Texans this off-season by his former team, The Cincinnati Bengals. The compensation for acquiring Mixon from the Bengals cost the Texans the 224th pick in the upcoming 2024 NFL draft, a 7th-round pick. In Mixon's last three seasons in Cincinnati, he started in every game but three and averaged 1395 yards from scrimmage over the three years. By most metrics, Mixon was a very talented running back, yet he was dealt for the least valuable capital to trade in the NFL: a seventh-round pick. Looking back to the 2022 season, when another noteworthy back was traded, we can observe an incredible discrepancy in trade value. Christian McCaffrey, 26 at the time of the trade, was dealt from the Carolina Panthers to the San Francisco 49ers in exchange for a 2023 second, third, and fourth-round pick and an additional 2024 fourth-round pick. In the three years leading up to the trade, McCaffrey played in 26 games (missing ten games in 2021 and 13 in 2020) and averaged 1189 yards from scrimmage in those three seasons. It is apparent that when on the field, McCaffrey is every bit as good as Mixon, but often goes the saying in the NFL, "the best ability is availability." However, in this trade, McCaffrey's injury history and age did not dissuade the 49ers from trading premier picks for him, but the age and career timeline for Mixon, who was only one year older, tanked his value. Even when comparing the contracts that both players would play on, McCaffrey is set to make about \$16,00,000 a season, whereas Mixon will only make about half of that at \$8,500,000 per season. Here is a case of two similarly aged running backs, past their initial rookie contracts and having had very

productive seasons. However, only one was valued highly, and it was the running back who had recently displayed durability concerns. At the same time, it is entirely possible to write these trades as anecdotal and not much to look into, but we believe this plays into the 2011 contract adjustment and the complex nature of football. While both running backs are very talented, only McCaffrey is known to be an almost equal receiver to his already impressive rushing ability. The NFL draft is intended not just to add new players but also to replace existing players. We propose that teams involved felt Mixon's services could be cheaply replaced in the draft, and due to this, he was dealt for low-value assets. In the case of McCaffrey, he was viewed as having an ability that is likely very difficult to replace in the draft, and he was then dealt for much more value despite his injury history. This case may serve as both an example and a reminder that the 2011 CBA did not just devalue the running back position; it made the times running backs have been able to differentiate their value from incoming rookies few and far between.

Closing Thoughts

Attempting to quantify changes in the sports world is often a difficult and, at times, futile challenge. Sports takes in factors far beyond what any empirical study can consider. Player emotion, team chemistry, fan interaction, leadership, etc, are all fundamental factors any analyst, player, or media personality will acknowledge, but all are impossible to quantify, let alone empirically test. However, we are confident in our approach to this challenge and findings. In a world where the running back position is losing value in terms of contract spending, there are an infinite amount of purely football explanations that can be used to attempt to explain this. Coaching, strength of schedule, and change in the college game are all valid ideas that explain how positional outlook has evolved. Testing all of these ideas however, would not only be extremely difficult, but it would not be possible under any reasonable time constraint, and they

are all factors that affect individual players differently. Nonetheless, the NFL contract by-laws are one of the few absolutes of the NFL. Every team and player must abide by the set contract rules, register for the draft to be eligible for the league, and field a running back who has to be paid through an NFL contract. All of our presented data have shown some reaction to 2011, while the explanation could hypothetically be purely football-related and have nothing to do with contracts. The likelihood of this occurring so suddenly all at the same time feels extremely implausible. Within the notes section below, we outline how previous research has affected our study, and at no point could we find any documentation that cites the 2011 CBA as the key reason for this devaluation. Some studies may mention the contract structure in a limited fashion, but we have yet to find a study that looks into contract structure as a determinant of positional value in depth. Again, it is impossible within the sports world to find an explanation for events with certainty. However, we have found through our analysis that the change to the rookie contract structure in 2011 is the likeliest explanation for the devaluation of the running back position and the reason behind major statistical shifts in production and positional salary changes.

Literature Review

PFF (2023)

The article states that the running back market has not grown in relation to the rest of the league, asserting that the top end of the running back market has seen a 12% growth since 2011,

but every other position group top end has seen a 60% growth in that same period. This is consistent with running back pay not running in proportion with the rest of the league. PFF (2023) focuses only on the top end of the market; however, this leaves much more room for research to be desired. Viewing only the top contracts is highly isolated and PFF (2023) even states that the contract of Christian McCaffrey (the current highest-paid running back in the NFL) was an outlier and would not be surpassed by other backs anytime soon. If this is, in fact, the case, then what good is viewing his contract if it is not indicative of what running backs are genuinely earning? In general, however, we believe that by only viewing the top contracts, one misses the ability to discern the value of a positional group. We say this because oftentimes, anyone at the absolute pinnacle of their position will be paid for their ability as a top-level player rather than paid in correspondence to deemed positional value. That being said, this is still our anecdotal opinion rather than a statistically backed assertion. This is what helped guide and inform my second set of data collection. In our research, we gathered the contracts of the thousand-yard runners and their corresponding contracts. In doing this, we captured a broader understanding of the running back market, which we felt was only partially given in PFF (2023). PFF (2023), after viewing the past decade's running backs who earned at least seven million dollars annually (this totaled 19, which raised sample size questions), compared their player grades for the first five years of their career to the last five years of their career. The problem with the presentation of the information is that their variables, such as "Rusher grade, stuff rate," among others, are arbitrary, and it is not fully clear how these values were calculated. PFF prides itself on its grades coming from "film analysis"; however, human error is inevitable given the nature of film analysis. PFF (2023), in the end, did not help serve our research as an informant so much as it served as a catalyst for questions that drove our research.

Eric Lofquist (2017). Does a Better Running Back Mean More Rushing? Game Theory and the NFL

There is a presumption that opposing defenses would be more adequately prepared for higher-quality running backs, thus lowering the rushing outputs of teams with better running backs. Through a fourteen-year NFL analysis of both player efficiencies and choice of play, Lofquist (2017) concluded that there was not, in fact, a negative correlation between rushing output and the quality of runners. This conclusion in and of itself is not that surprising to us, given that the only way to be deemed a "top rusher" is generally through top-of-the-league rushing outputs. The methods by which Lofquist (2017) came to this conclusion were largely based on the game theory of the value of running the ball as a function of having a talented running back versus defenses being better prepared to stop the run against better backs.

However, while talented backs were vital to successful run games, offensive coordinators and their philosophy are often a more prominent driver of running back usage, and the back's talent would serve as a multiplier to the production. A coordinator at the NFL level who prefers to run the ball will run the ball regardless of who is in the backfield, and the lion's share of defensive adjustment is founded on team play calling rather than the player itself. Nonetheless, to become a top rusher in the NFL, one must achieve top rushing statistics, and whether or not this indicates play-calling is irrelevant. This is irrelevant because, whether or not it is due to playcalling, there has been enough scheming to make a rusher at the top of the league; thus, further research into the coordinator's role becomes non-valuable.

Lofquist (2017) asserted a running back rating, explained as a starter's ability over a replacement level player, as an independent variable, with the pass play percentage as the dependent variable. This may not be informative of a player, however, because there are far too many factors in a player to reliably track how the quality of runner impacts run or pass playcalling, especially given that run or pass is often endogenous not only to the game situation but also the pass-catching capabilities of the player. Lofquist (2017) also used a series of case studies following specific players and the team's success once the star running back went down, which are too anecdotal to be reliable to be utilized in my research. For the research we will be conducting, this study served well in terms of granting surface-level ideas that we would like to look into researching further. For example, the replacement value of the running back position is something that we have gone into data research of; however, we will be collecting this data in mass and only using individual case studies as anecdotal supplementation of a more significant statistically-based idea.

Chase Kennedy (2010). Age vs Wear And Tear: The Deterioration Of An NFL Running Back

The effects of both age and workload of NFL running backs on their on-field production. Interestingly, Kennedy (2010) finds no clear baseline for a standard age for production drop-off in running backs, but they did find a workload amount that suggested a coming production drop-off. The declining trend of running backs earning their second contract could be based on usage rather than age; however, while we found this study's premise valuable, we are led to believe the declining value lies within the ease of replacement. Unlike Lofquist (2017), we do not refer to replacement as an active roster backup but as undrafted rookie running backs who

will become available in the upcoming draft. Viewing the rate of rookie running back success relative to veteran success could serve as a major insight into the value of running backs. Not only does this apply to running backs, but every offensive skill position could also add comparability to discover actual positional value based on ease of replacement.

Matthew Kraeutler et al (2018). What is the Value of a National Football League Draft Pick?

Kraeutler (2018) explores the value of each draft pick by round, most interestingly weighing the value of rookie contracts against predicted pick success. To measure predicted success, Kraeutler used data from the first three rounds of three draft classes: 2010, 2011, and 2012. Kraeutler created a value statistic for the sample player pool and used this value statistic to determine average scores for round 1 players sorted into "group A" and players drafted in rounds 2 and 3 as "group B." How the "value statistic" was created became a point of contention within the validity of certain conclusions made in the study. The "value statistic" is formed by collecting position-specific stats of drafted players in their rookie years and dividing the combined on-field performance score by the drafted player's salary cap hit. This formed a rookie performance per dollar metric, which the study used to predict how valuable a first-round pick was as opposed to a second or third-rounder. In addition to this, combined measurements by position group were taken to see how players of varying combined scores were available throughout the three rounds. The problem with grading players on this created measurement of value is its incredibly arbitrary nature. It requires the assumption that using performance per dollar, where dollars are a linearly paid value for statistical output, is accurate. In reality, contract rules such as minimum salaries or baseline statistics for an average player would make this linear metric less valuable. For example, if a league minimum of defensive linemen is expected to generate "x" amount of sacks

per year. Paying for an all-pro would not be for them to get a different variable "y" sacks, but instead paying them for how much greater than "x" or a minimum-level player they are. Dividing every stat by contract heavily favors players who are paid less because it cannot begin to weigh the benefit of certain first-round players whose contributions extend beyond stats. That being said, the premise is that second-round players gain added value because of their lower contracts and when weighing contract terms against production. The value of later rounds shoots up relative to first-round players, not necessarily to exceed the value of the first round, but the idea of drafted player value from a contractual and salary cap perspective rather than only a statistical output perspective is important for our research. The point of how rounds impacted high-performing combined players' availability was exciting but of little value for the research done in this paper. Unsurprisingly, the paper found in both statistics and availability of high combine performers to be higher in first-round players and taper off to the subsequent rounds. However, Kraeutler (2018) also found rounds 2 and 3 to be of more value than rounds 1, given his metrics. Again, as previously mentioned, this is not the entire case, and the metric created cannot come close to consistently grasping the value of how a player contributes on the field. Their ideas were still of great use and helped drive the research done in this thesis.

Aryan Von Eicken (2017). An Analysis of the Efficiency of the NFL Draft

Von Eicken (2017) helped drive our study regarding the author's research around positional value in the draft. The paper contains little substance in terms of contractual or salary cap analysis. However, the analysis conducted still served this thesis very well. The author confirmed the idea of first-round players outperforming players drafted in the ensuing rounds and found that some positions' available value was maintained throughout the draft better than

others. The authors used a metric created by the sports statistics website *Pro Football Reference*, called CarAV, to gauge how players' careers will pan out. We have some reservations about any metric that attempts to amalgamate field statistics into one value, as mentioned in Kraeutler (2018). CarAv does a better job of factoring in more statistics (including statistics of team performance, not just individual) than Kraeutler (2018). Also, by not attempting to weigh CarAv against contracts or salary cap hits, Von Eiken (2017) removes the arbitrary nature of how a dollar spent should scale relative to statistical output. The author splits the draft into rounds 1, 2, 3, 4, 5, and 6, 7. In doing a 2-sample T-test, the author found differences in a quarterback taken in the first as opposed to any other round but found no evidence to support that second & third-round quarterbacks would go on to have careers more productive than those taken in the sixth and seventh rounds. This raised interesting conversation within this thesis about running backs holding more value in earlier rounds because of the relative confidence that a running back draft selection will yield more outstanding production. Von Eicken (2017) attempted to correct the skew towards earlier drafted players getting more starting opportunities, given that CarAv can only be accumulated through games. In order to do this, a switch was made from CarAv to CarAv per game; this correction feels unnecessary because first-round players earned the entitlement to be in a more contributing position. Throughout even one season, players will play more or less depending on production quickly, thus correcting a career timeline. The finding that aided the research done in this thesis was that of all positions, running backs were amongst the ones providing the most value in the first round, and that it was other positions, such as WR, whose values could be more easily found in later rounds. Von Eicken (2017) study aided the research and supported assertions that running backs aren't devalued as a position; rather, they are devalued past their rookie deal due to existing NFL contract rules and structures.

Gina Nguyen (2022). What Is The Average Length of a NFL Career, By Position?

Nguyen (2022), in this relatively short research article, presented the career average in terms of years on an NFL roster by position as of 2022. The data within the article was not presented in an easily understandable way regarding how these averages were found. The authors state these career averages to be of the 2022 season. However, how the career length was calculated needs to be clarified. While time on a roster can be easily understood as the primary metric here, what exactly forms the sample size needs to be clarified. Was this calculated as the average career length of every NFL player up to this point or of the players whose careers ended in 2022? This was the average career length by positional breakdown. Nevertheless, even with the flaws of this article, we were still able to take small anecdotes, which aided in how the research in this thesis was framed. Nguyen (2024) finds that the average career in the NFL sits at 3.3 years. More importantly, only Kicker and Quarterback exceed an average career length of 4 years, at 4.87 and 4.44 years, respectively. Going further, only offensive and defensive linemen have average career lengths above 3, with 3.63 and 3.24, respectively. Taking this information at face value supports our direction away from not viewing by career timeline in extensive depth. Every year in the NFL, at least 224 players are drafted, and undrafted free agents only add to this number. Rookie contracts are locked on a four-year timeline and are set in contract value regardless of position. There is little value in investigating the average career length because the averages are skewed so far down by players who would never pan out in the NFL, regardless of position. Looking at how positions are valued regarding salary caps, spending would come in the second contract or where they are drafted. Average career length would only be beneficial regarding age, where we can see how long positions will typically last, which was already done in this thesis. However, when viewing the average ages of productive players, this was not done

with a sample size of every player due to skewing of players who were never going to make it in the NFL, but instead of productive players who make rosters, what are the average ages of these players?

Barnwell (2023), Why Star NFL running backs have been devalued: What's next?

Barnwell (2023) cites six reasons for devaluing the running back position, specifically star running backs. His first line of reasoning is that the league has moved toward a more pass-heavy direction, citing the 2007 Patriots as the catalyst for the shift to passing. This claim is unfounded in significant statistical evidence, and we are unsure on what basis or metric Barnwell (2023) is stating this. Table 4 shows a league average of 33.3 passing attempts per game, only down 0.4 from the 33.7 average attempts in 2023. Looking into the past decade, 1997 the passing attempts per game sat at 32.8. In 26 years, the NFL is only averaging a single passing attempt more, which is hardly enough to assert that a spike in passing has driven the running back market down. While Barnwell (2023) could be referencing a spike in passing yardage rather than passing attempts. Average passing yards in the past 30 years have only increased about 5 yards from 213.6 in 1994, the first year of the salary cap, to 218.9 in 2023. Within this thesis, we make significant efforts to reject the notion that an influx of passing has driven running back value down, and thus, we cannot support Barnwell (2023) in this claim.

The second reason for runningback devaluation made by Barnwell (2023) is the quarterback's involvement in the running game. Barnwell has a much stronger case here than the previous assertion; however, empirically and theoretically, this argument comes up short.

Empirically, when viewing Figure 18²⁸ and 19²⁹, which uses the same 150 scrimmage leaders from 1994 data; we see growth in quarterback rushing. We see in Figure 18 the chart displaying quarterbacks' share of top-performing scrimmage yards generators, and Figure 19 shows the # of quarterbacks who finish in the top 150. Observing both graphs move almost perfectly in sync, but this also tells us that quarterback rushing is driven by a handful of quarterbacks each year. That being said, quarterbacks who finish outside the top 150 may still be more productive rushers on average. However, Only 16 quarterbacks, half of the league's starters, finished this season with over 200 rushing yards. We can infer that half of the league at least not only does not have a rushing threat at quarterback but would thus still need their rushing offense to come from the running back position. Rushing quarterbacks may be on the rise, but they are not something the league has not seen before; they are not figures 18 and 19 in 2000. It is for this reason, on top of the fact that even with this influx, many teams still do not possess a rushing quarterback, we can safely reject Barnwell's (2023) second explanation for running back devaluation.

²⁸ Figure 18 uses a sample group previously used in this thesis, the top 150 leaders in NFL yards from scrimmage. The x-axis is years/seasons, with the y-axis serving as the % of yards generated by the top 150 yards from scrimmage leaders attributable to quarterbacks.

²⁹ Figure 19 uses the same sample group as Figure 18, with the same x-axis, except the y-axis is not represented as the # of quarterbacks who qualified for the 150 yards from scrimmage leaders by season.

Figure 18

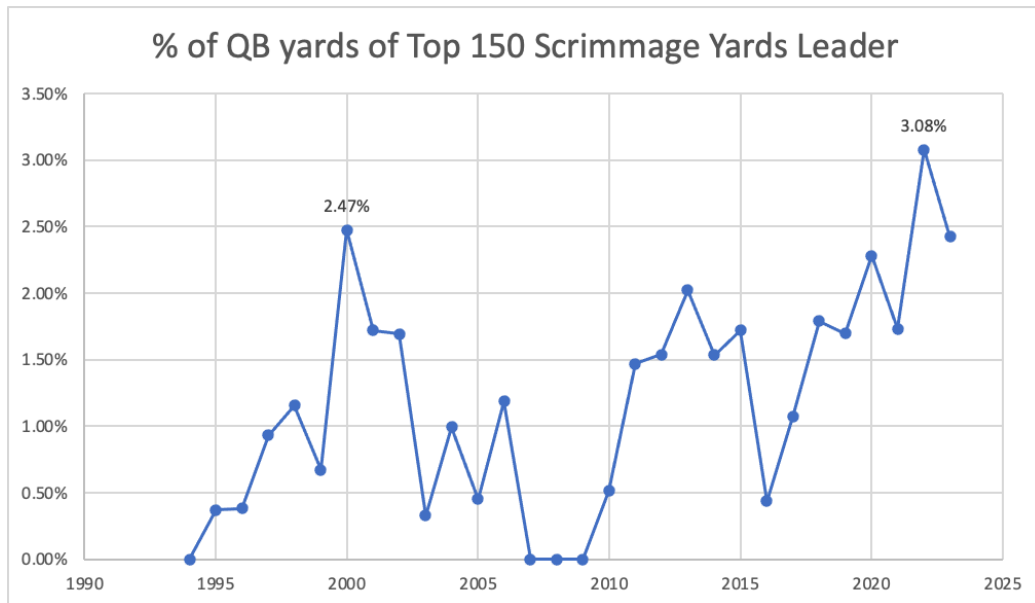
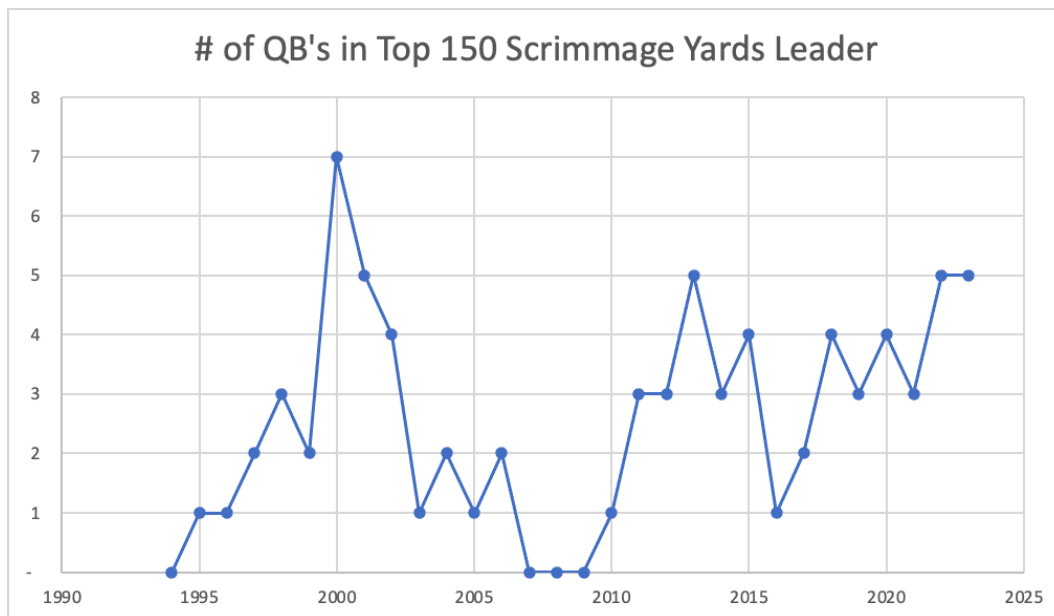


Figure 19



Barnwell (2023) uses the "running backs by committee" argument to explain why running backs have become devalued. This is a point with which we agree; however, within his reasoning, Barnwell (2023) makes no note of why this was even possible. In this research, we

found the 2011 CBA change to be why running back by committee is a viable roster-building tactic, whereas Barnwell (2023) provides rushing statistics for rushing totals of lead-backs lowering over time. While we do agree with Barnwell, his reasoning for this explanation does not fully align with our explanation of the causation of running back devaluation.

Barnwell (2023) cites the 2011 CBA draft changes as the fourth reason, which is the crux of our research here; however, he completely misses the mark in his explanation. Barnwell (2023) explains the 2011 CBA change effects as changes to positional value based on the ability to acquire premier positions, namely quarterbacks, on a discount due to the cost-controlled nature of rookie contracts. There is some merit in the value of positions, like quarterbacks being able to be signed under market value on their rookie deals. This only applies to teams that are in need of a quarterback, which at most is four teams in the draft. The benefit of the rookie quarterback deal would be to pay a premium in free agency for positions a team has not drafted highly or drafted well, such as running back. Examples include the 49ers signing Christian McCaffrey while Brock Purdy is on his rookie deal and the Texans trading for Joe Mixon while CJ Stroud is on his rookie contract. Quarterbacks drafted in the first round, which is not a new phenomenon in the post-2011 era, could not effectively be proven by Barnwell's (2023) data and argument.

Barnwell notes poor running back drafts for the years of 2006-2010. This is an argument that is difficult to engage within the bounds of this research, as the reasoning provided by Barnwell (2023) is anecdotal and not primarily based on statistics. For this argument, Barnwell disregards the top backs of this period, which include fifth all-time leading rusher Adrian Peterson, as being too few and then throws out a handful of names as being poor players. He does not provide any metric in which he deemed these players poor; one could assume it is based

on their career-rushing outputs, but this argument generally feels too unstructured to be engaged within the bounds of a research essay.

The sixth and final reason for running back devaluation provided by Barnwell (2023) would be the incorporation of advanced analytics revealing more accurate value measurements of players to teams. Advanced analytics has its place in the sports world. A sport such as football incorporates so many difficult-to-track and measure factors into team and player performance alike. Advanced statistics can sometimes lose their value in explaining on-field occurrences because there are often too many factors at play. However, in general, we accept this premise by Barnwell, but we also offer alternatively that advanced statistics may be vital in driving current running back value perceptions. This was largely possible due to the 2011 CBA change, which gave most of the leverage in value negotiations to the front offices, out of the hands of the running backs.

Overall, while we disagree with much of Barnwell's (2023) argument, his reasoning and explanations provided much discussion and analysis for this research. We could find and support our explanation for running back devaluation mainly by analyzing and rejecting some of his explanations.

OpenAi, (2024). In one paragraph please explain why running backs have been devalued in the past 20 years?

In a more avant-garde way to gather the popular sentiments for explaining the case of running back devaluation, we asked ChatGPT what it thought about the question. It responded, "Running backs have been devalued in the past 20 years primarily due to shifts in offensive strategies and the evolution of the passing game in the NFL. With rule changes favoring passing

offenses, teams increasingly rely on dynamic quarterbacks and prolific receivers to move the ball down the field efficiently. Additionally, the emphasis on player safety has led to a greater focus on preserving the health and longevity of running backs, who historically endure high physical punishment. Consequently, teams have shifted towards utilizing running back committees to spread the workload and minimize the risk of injury, leading to a perception of decreased value for individual running backs in the modern game."

ChatGPT could not offer any new explanation or line of thought for explaining the question. We cannot say we are too surprised as the chatbot draws upon existing information on the web, but the hope was that it could find an answer somewhere beyond the scope of our research. Nevertheless, the points OpenAi (2024) provided are contested mainly, disproven, and or alternatively explained within the thesis. Both shifts in strategy and the evolution of passing are only partially supported by the statistics outside of an uptick in completion percentage. Interestingly, the changes may have given the appearance of a more passing-favored game; the passing output has not shifted much in the past 20 seasons. We disagree that player safety was a driver in creating the running back by committee strategy. Rather, we attribute this to the 2011 CBA change, as mentioned throughout the thesis. While ChatGPT was a fun idea in concept, it would largely and unsurprisingly be a regurgitation of existing ideas.

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